

Tomahawk 17315



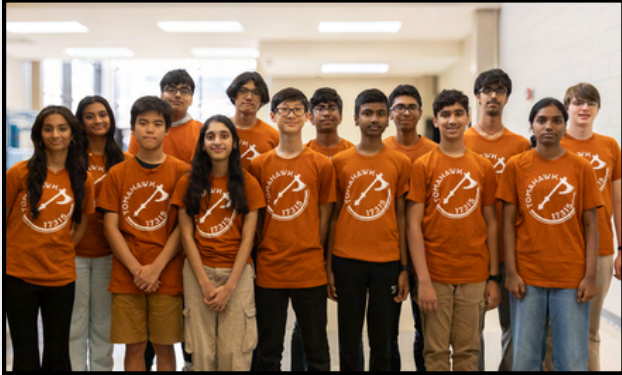
Outreach Portfolio

2025-2026





OUR TEAM



We are team #17315, **Tomahawk**, a robotics team from Westwood High School. Our team is made up of a mix of **rookies** and **returning members**, all excited to dive into the world of robotics together. Each teammate brings **unique traits**, whether in designing, coding, building, or so much more. With fresh ideas and determination, we aim to create a robot we're proud of while also having fun with this year's Decode game!

Outreach GOAL

To lower the barrier of entry for robotics and STEM, making robotics and success more accessible for all students.

Reasoning: This year, from our meetings with professionals, to our interactions with our local community and abroad, **we wanted one common goal and theme for this years outreach**; a final product and **lasting impact** that all of this work would help create. Together, from our early year connect meetings and some of our past experiences on other teams, we set a new goal that we hoped to achieve this year through all of our efforts. We **chose this specific goal** because of our past experiences, that **when we all started FIRST, many of us discovered robotics very late** and also from our experience, we realized even then, **we didn't have much help or resources to help us learn fast.**

The Outreach Team

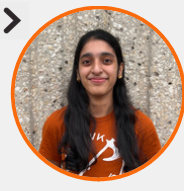


Ritvij S.

5 Years in FTC

About Me

My passion for STEM and hands-on engineering led me to join FTC Team 17315 Tomahawk, where I've grown creatively and collaboratively. In the future I hope to be a Mechanical or Electrical Engineer. In my free time I like to go camping and spend time with my friends.

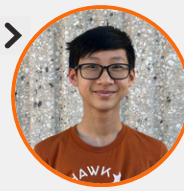


Prahastha P.

2 Years in FTC

About Me

I joined FTC Tomahawk to explore my passion for STEM and teamwork. being a part of this team helped me grow my problem solving skills and learn how engineering and collaboration go hand in hand.

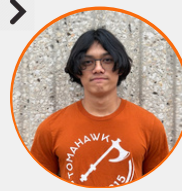


Isaac K.

2 Years in FTC

About Me

When I joined, I wanted to learn about how robots work, and how to design a functional product and I was not disappointed! Now, I work to learn even more, and spread everything I've learned to as many other people as I can.



Jusuf R.

2 Years in FTC

About Me

I joined Tomahawk because of its ambition, drive, and commitment to making a real impact on our community and underprivileged communities too. Outside of robotics, I like to play the viola and ride my bike.



Daniel L.

2 Years in FTC

About Me

I joined Tomahawk because it is a great place where we get to connect, work together, and solve challenges. By being part of this team I hope to pursue my passion for robotics and STEM. I also like to play the violin and sleep.



Jay P.

2 Years in FTC

About Me

I joined Tomahawk because I enjoy competitive environments where I can showcase my talents the best. During my free time, I like to hang out with friends and play competitive chess.



Mentors

Josh E.



Connection:

-FRC 2583 alumni, volunteer for the school robotics club

Contribution:

Hardware mentor, rules analyst who helped us strategize.

Rodelio C.



Connection:

- Former Robotics class teacher

Contribution:

Helped connect us with the school we hosted Project Philippines.

Santosh C.



Connection:

-Former 3-year head coach of 19853 Roboteers, Microsoft Engineer

Contribution:

Assisted with outreach, inspired BDRP, and provided portfolio feedback as a 3-year veteran judge.

Amit U.



Connection:

-Former 20240 coach

Contribution:

Advice on materials for robot construction as well as guided us through the manufacturing process for cutting certain items.

Youth mentors

Satvik C.



Connection:

-17315 Tomahawk alumni, 4 years in FTC

Contribution:

Guided our robot-building approach and helped members learn hardware and CAD.

Shubh D.



Connection:

-**21323a**, 2583 FRC Software Lead

Contribution:

Helped us **debug** our robot and overall ensure that our code **consistently** worked

Demian M.



Connection:

- 20716 alumni, 19746 alumni, 5 years in FTC

- Into the Deep Winning Alliance at Worlds

Contribution:

Advised on Software Tuning, strategic planning, smart decision-making, and time management for an effective season.

Avi S.



Connection:

- 30030 Exodus member, 2 years in FTC

Contribution:

Student mentor who advised on design, mechanisms, Onshape, and gave recommendations.

Nipun N.



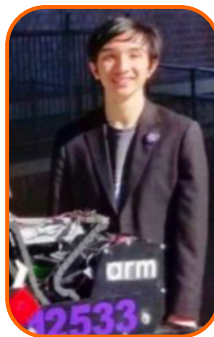
Connection:

-**6210** alumni, Cornell University Student

Contribution:

Assisted us on how to effectively prepare for presentation as well as how to efficiently approach the software side of portfolio. Also was a software mentor helping us debug our code.

Peter L.



Connection:

12533 FTC former captain and alumni, former 19746 mentor

Contribution:

Gave us analysis advice as well as both software and general bot designing advice. Also told us how to network for alliance selection.

Angela Z.



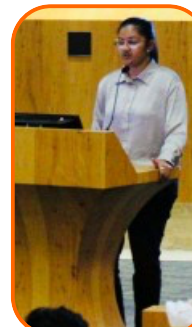
Connection:

-4545 Ouroboros alumni

Contribution:

Assisted with outreach, **networking** strategies, and **presentation** preparation.

Anvitha K.



Connection:

-18030 captain **alumni**, UC Berkeley School of business

Contribution:

Assisted with **presentation**, outreach ideas, and **pits** preparation to ensure we could **meet the judges standards** for different award categories.

Proof of Contact

Although we established which companies we wanted to contact, we realized that it was difficult to complete the process. However, we eagerly continued to contact as many companies as possible and we were able to successfully meet with a variety of professionals. Below we have kept record of all our company interactions.



CONTACTED



SCHEDULING



MET WITH

12

02

18



PROOF OF CONTACT: 06/13/25

potential meeting Inbox xTomahawk FTC <westwoodtomahawk17315@gmail.com>
to angela.rizzi

Dear Ms. Rizzi,

We are FTC team 17315 Westwood Tomahawk and we are from Austin, Texas. As a FIRST robotics team, we are developing as roboticists. Thus, we were wondering if we could potentially schedule a meeting with you to learn more about your background, your job, and the design process used at NASA.

Thanks,

Tomahawk 17315

Address: 12400 Mellow Meadow Dr, Austin, TX 78750

Gmail: westwoodtomahawk17315@gmail.comYouTube: www.youtube.com/@tomahawk17315Team Website: tomahawkFTC.comOrganization Website: westwoodrobots.org/FTC

PROOF OF CONTACT: 6/14/25

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to knewman, kj, knewman, kj, knewman

Hello There! My name is Ritvij Sharma and I am writing on behalf of my FTC team 17315 Tomahawk. We were hoping to connect with you to attempt to problem solve technical issues and debug tasks that come as being an aerospace hardware engineer, and skills more. Thanks,

Tomahawk 17315

Address: 12400 Mellow Meadow Dr, Austin, TX 78750

Gmail: westwoodtomahawk17315@gmail.comYouTube: [youtube.com/@tomahawk17315](https://www.youtube.com/@tomahawk17315)Team Website: tomahawkFTC.comOrganization Website: westwoodrobots.org/FTC

PROOF OF CONTACT: 06/14/25

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to hsinlin.lai, hsinlin.lai

My name is Ritvij Sharma and I am speaking on behalf of my FTC team 17315 Tomahawk. We were wondering if you could also meet with us, potentially in person. As part of being a FIRST robotics team, we are developing as roboticists. Thus, we were wondering if we could potentially schedule a meeting with you to learn more about your background, your job, and the design process used at HP.

Thanks,

Tomahawk 17315

Address: 12400 Mellow Meadow Dr, Austin, TX 78750

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PROOF OF CONTACT: 06/15/25

Potential Visit

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to tmi

Hello there! My name is Ritvij Sharma and I am speaking on behalf of 17315 Westwood Tomahawk. We were hoping to connect with you to attempt to problem solve technical issues and debug tasks that come as being an aerospace hardware engineer, and skills more. Thanks,

Tomahawk 17315

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Proof of Contact



College of Engineering

PROOF OF CONTACT: 06/14/25

PROOF OF CONTACT: 06/13/25

Potential Meeting



Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to r Lambert, lambsoftwork, lambsoft99 ▾

Hello there! I am Ritvij Sharma writing on behalf of 17315 Westwood Tomahawk. We were wondering if we would be able to meet with you if we could learn about the real life applications of robotics, troubleshooting techniques used by you in Salesforce, and any other advice you have for High Schoolers aspiring to join the STEM field in the future. Thanks,
Tomahawk 17315

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Team Website: tomahawkFTC.com
Organization Website: westwoodrobots.org/FTC

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to pranav.r.srivastava ▾

Dear Pranav,

My name is Ritvij Sharma and I am speaking on behalf of 17315 Tomahawk. I was wondering if we could learn about the real life applications of robotics, troubleshooting techniques used by you in Salesforce, and any other advice you have for High Schoolers aspiring to join the STEM field in the future. Thus, we feel your advice would be greatly appreciated. Thanks,
Tomahawk 17315

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Team Website: tomahawkFTC.com
Organization Website: westwoodrobots.org/FTC



PROOF OF CONTACT: 07/05/25



PROOF OF CONTACT: 06/14/25



Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to sarvesh.bharadwaj, sbharadwaj, sar123 ▾

Dear Mr. Bharadwaj,

My name is Ritvij Sharma and I am speaking on behalf of my FTC robotics team 17315 Tomahawk. We were wondering if we would be able to meet with you if we could learn about the real life applications of robotics, troubleshooting techniques used by you in Salesforce, and any other advice you have for High Schoolers aspiring to join the STEM field in the future. Thanks,
Tomahawk 17315

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Team Website: tomahawkFTC.com
Organization Website: westwoodrobots.org/FTC

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to Ashish ▾

Hello! I have signed up for the time slot of 5:30 on July 26th but haven't documented my experience yet. Please feel free to send over any additional advice you have for High Schoolers aspiring to join the STEM field in the future. Thanks,
ritvij.sharma10@gmail.com if they need to reach me.

Thanks,
Tomahawk

TEXAS Robotics

PROOF OF CONTACT: 06/09/25



The University of Texas at Austin
Texas Inventionworks
Cockrell School of Engineering

PROOF OF CONTACT: 06/20/25

Invitation from an unknown sender: Ritvij Sharma and TX Robotics
11:30am (CDT) (ritvij.sharma10@gmail.com) [Inbox x](#)



TX Robotics Social <txroboticssocial@gmail.com>
to me ▾

This event isn't in your calendar yet You haven't interacted with txroboticssocial@gmail.com before. Do you want to add it to your calendar?

Add to calendar

Report spam

Jul

Ritvij Sharma and TX Robotics

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to scott.evans, benjamin.black, audrey.boklage, s.ferraro, jjohnson, steven.kim, tiw-support ▾

Hello! My name is Ritvij Sharma and I am speaking on behalf of FTC team 17315 Tomahawk. We were wondering if we would be able to meet with you if we could learn about the real life applications of robotics, troubleshooting techniques used by you in Salesforce, and any other advice you have for High Schoolers aspiring to join the STEM field in the future. Thus, we feel your advice would be greatly appreciated. Thanks,
Tomahawk 17315

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Team Website: tomahawkFTC.com



Proof of Contact



Microsoft

PROOF OF CONTACT: 06/25/25

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to praks78 ▾

Dear Mr. Prakash,

We are team 17315 Tomahawk and we were interested in learning about your companies and the strategies you implement to work together.

Thanks,

Tomahawk 17315



PROOF OF CONTACT: 07/13/25

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to contact ▾

Hello! We are a team of robotics students. As part of being a member of the longhorn racing facilities in Austin and learn perhaps more about your facilities.

Thanks,

Tomahawk 17315

facebook

PROOF OF CONTACT: 07/03/25

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to asondhi.misc ▾

Dear Mr. Sondhi,

My name is Ritvij Sharma and I am writing on the behalf of my team to potentially schedule a meeting to learn about the responsibilities of a chip designer and responsibilities in chip design.

Thanks,

Tomahawk 17315

micron®

PROOF OF CONTACT: 07/12/25

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to Priya.girinath ▾

Dear Ms. Girinathan,

My name is Ritvij Sharma and I am speaking on the behalf of my team and potentially any advice you may have for high schoolers. As a part of our team, we feel scheduling a meeting with you would not only heighten our knowledge but also provide us with valuable insights.

Thanks,

Tomahawk 17315



tenstorrent

PROOF OF CONTACT: 06/09/25

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to Rajeev ▾

Dear Mr. Srivastava,

My name is Ritvij Sharma and I am speaking on the behalf of my team to learn from STEM industry professionals to learn from their experiences. Thus, we are aspiring engineers in the STEM industry.

Thanks,

Tomahawk 17315



Stanford
University

PROOF OF CONTACT: 06/20/25

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to arajendran ▾

Dear Mrs. Rajendran,

My name is Ritvij Sharma and I am speaking on the behalf of my team and potentially any advice you may have for us.

Thanks,

Tomahawk 17315

Proof of Contact

SAMSUNG

PROOF OF CONTACT: 07/11/25

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to j.sabrsula ▾

Dear Ms. Sabrsula,

My name is Ritvij Sharma and I am v
STEM industry in our community and feel that visiting th
Thanks,
Tomahawk 17315

HUSK.

PROOF OF CONTACT: 07/13/25

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to info ▾

Dear Husk,

My name is Ritvij Sharma and I am writing to you
about STEM applications in the real world. We also feel that
to potentially schedule a meeting with a representative from
Thanks,
Tomahawk 17315



Cornell University.

PROOF OF CONTACT: 08/21/25

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to nipun.nagendra9 ▾

Dear Nipun,

My name is Ritvij Sharma and I am speaki
and would really like to listen to any advice you may ha
Thanks,
Tomahawk 17315

TESLA

PROOF OF CONTACT: 07/12/25

Tesla Factory Tour Aug 5

Arjun Averineni <arjun.averineni123@gmail.com>
to santosh ▾ 10:22 AM (0 n

Dear Mr. Yerramreddy,

My name is Arjun Averineni, and I'm reaching out on behalf of FTC Team 17315 Tomahawk. We are a robotics team consisting
the First Tech Challenge competition annually.

This season, our team is eager to explore real-world engineering, design, and production methods, as well as automation, and
we may implement these methods in our team for improvement.

With that in mind, I wanted to ask if there might be any availability for a Tesla Factory Tour on August 5th for 7-8 people.

Please let me know if that date might be feasible or if there are any other possible options. Thank you for your time!

Sincerely,
Arjun Averineni
FTC Team 17315 - Tomahawk



PROOF OF CONTACT: 10/09/25

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to alexandernuccitelli ▾

Dear Mr. Nuccitelli,

We are Westwood Tomahawk 17315. We were wondering if we would be able to potentially s
have for high schoolers. Nvidia is a well known company and we were hoping you could give us some insight ab
experiences at Nvidia.

Thanks,
Tomahawk 17315

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Team Website: tomahawkFTC.com
Organization Website: westwoodrobots.org/FTC

indeed

PROOF OF CONTACT: 08/01/25

Tomahawk FTC <westwoodtomahawk17315@gmail.com>
to Tim ▾

Dear Mr. Kao,

We are an FTC Robotics team 17315 and we w
would really prefer an in person meeting due to it being more
from,
Tomahawk 17315

Proof of Contact

	 TEXAS A&M UNIVERSITY Engineering
PROOF OF CONTACT: 10/14/25	PROOF OF CONTACT: 09/14/25
Tomahawk FTC <westwoodtomahawk17315@gmail.com> to john.hwang@gmail.com Hello Mr. Hwang, My name is Ritvij Sharma and I am speaking on behalf of 17315 Tomahawk. I was wondering if we could connect and learn many companies such as SideStep. We feel that your advice from your experiences from working in the industry to making your own company. Thanks, Tomahawk 17315 Address: 12400 Mellow Meadow Dr, Austin, TX 78750 Gmail: westwoodtomahawk17315@gmail.com YouTube: youtube.com/@tomahawk17315 Team Website: tomahawkFTC.com Organization Website: westwoodrobots.org/FTC 	Dear Abraham, My name is Ritvij Sharma and I am speaking on behalf of 17315 Tomahawk. I was wondering if we could connect and a former Tomahawk alumni who ranked among the better teams in the 2022-2023 Powerplay season and helped lead Tomahawk your advice would be greatly appreciated and would help us in our upcoming season. Thanks, Tomahawk 17315 Address: 12400 Mellow Meadow Dr, Austin, TX 78750 Gmail: westwoodtomahawk17315@gmail.com YouTube: youtube.com/@tomahawk17315 Team Website: tomahawkFTC.com Organization Website: westwoodrobots.org/FTC 
	
PROOF OF CONTACT: 11/17/25	PROOF OF CONTACT: 12/15/25
Tomahawk FTC <westwoodtomahawk17315@gmail.com> to avikacker@gmail.com Hello Mr. Kacker, My name is Ritvij Sharma and I am speaking on behalf of Westwood Tomahawk. We were wondering if we would be able to your field and potentially any advice you may have for high schoolers. As a part of FTC robotics, one of our main goals is to learn more about the community. Thus, we feel scheduling a meeting with you would not only heighten our knowledge of a certain STEM field, but could help us in the Thanks, Tomahawk 17315 Address: 12400 Mellow Meadow Dr, Austin, TX 78750 Gmail: westwoodtomahawk17315@gmail.com YouTube: youtube.com/@tomahawk17315 Team Website: tomahawkFTC.com Organization Website: westwoodrobots.org/FTC 	Tomahawk FTC <westwoodtomahawk17315@gmail.com> to DavidVinnik10 Dear Mr. Vinnik, We are Tomahawk 17315 and we are an FTC team who competes in the GEMS league. We experience at Bae Systems. I believe that this meeting would be really beneficial to our learning too. Thanks, Tomahawk 17315 Address: 12400 Mellow Meadow Dr, Austin, TX 78750 Gmail: westwoodtomahawk17315@gmail.com YouTube: youtube.com/@tomahawk17315 Team Website: tomahawkFTC.com Organization Website: westwoodrobots.org/FTC 

Overall Connect Results



**Industry
Professionals
met**



**Industry
Standard
Facilities Toured**



**University
students and
alumnis met**



CONNECTING

with professionals

Rebel Robotics - 18139

REBEL ROBOTICS



Goals:

Gain insightful advice on awards and portfolio for the upcoming decode season from an accomplished team

WHO:

Caleb and Josie, members of Rebel Robotics, winners of the *Inspire Award* at the First Tech Challenge *World Championship* in Houston

TEAM TAKEAWAYS:

Ritvij - "Meeting with Rebel robotics, the team that had won the Inspire award at worlds was a great experience. We learned to keep **interview answers concise**. We also learned the most optimal pit design would be to have the robot in the center with visuals on the side, letting judges explore and ask better questions."

Deekshita - "Rebel Robotics taught us how to **efficiently attract sponsors to further our cause** and how to unite all our outreach initiatives towards a single objective to create a sort of story to better showcase our efforts to the judges."

Prahastha - "Rebel Robotics helped me see how important it is to **tell our story in a clear and engaging way**. They showed us how a strong intro and confident delivery can really make people remember our team. they also recommended that since it was still very early in the off-season, that we should identify a main goal we wanted to pursue and make it so all of our outreach we pursued centered around that"

Jusuf - "Meeting with Rebel Robotics gave us a **better understanding of how to approach sponsorships**. They taught us the importance of **building connections through conversation before making a formal request** and emphasized the importance of preparing early."

CONNECTING

with professionals

SAP



Goals:

Learn more about the job of a product manager at SAP, their responsibilities, their strategies used to overcome issues, and the application of this to FTC

WHO:

Purnima Srivastava, the **VP of Product Management** at SAP

TEAM TAKEAWAYS:

Ritvij - "Mrs. Srivastava explained how planning is 80% of the work SAP does while only 20% is the actual. One way I feel we could **apply this to our own FTC team** is to **have a thorough plan from the beginning of the year to need to make less adjustments** to complete the objective and instead spend more of that **time optimizing** what we already have and designing better scoring ideas with no pressure from the timings of the competitions."

Prahastha - "It was an honor to meet Mrs. Srivastava. She shared valuable tips on marketing and planning that will definitely help in our upcoming robotics season. **She showed us how important it is to identify the right problem and plan thoroughly before executing.** Her insights on approaching new challenges with a **clear strategy** were inspiring and eye-opening. I'm grateful for the opportunity to learn from her."

Arjun - "Meeting with Mrs. Srivastava was a great experience. From her we learned that SAP **relies heavily on team feedback and customer input** to shape its products. From her, we realized that even **professionals depend on diverse perspectives.** As an FTC team, I feel that we'll benefit from **involving everyone in design and build decisions.** Whether it's brainstorming a new intake or testing autonomous code, I feel more thorough **team discussions can help us create smarter, more reliable solutions.** Everyone's voice should be heard."

CONNECTING

with professionals

Axon



Goals:

Gain insightful advice on how to overcome adversity, the current state of the software field, and advice from industry professionals to aspiring software engineers

WHO:

Gbenga Badipe, *Principal Technical Program Manager* at Axon

TEAM TAKEAWAYS:

Ritvij - "It was great meeting with Mr. Badipe. I learned the importance of **implementing feedback and leading with a focus on the bigger picture**. He emphasized realigning the team when needed and sticking to core principles during challenges. By prioritizing effectively and avoiding low-impact distractions, leaders can keep their teams motivated and on track."

Jusuf - "From our meeting with Mr. Badipe, I learned that **strong leadership balances structured communication with genuine human connection**. While async messaging is efficient—something our team already uses—**regular calls are key for clarity**. Mr. Badipe also emphasized **building relationships beyond work to prevent burnout and foster team culture**. Clear, transparent communication around changes keeps everyone aligned and engaged."

Arjun - "Mr. Badipe said something in our meeting that really stuck with me: **"Live through the experiences of your target audience so you can serve their needs."** That idea deeply resonated with me. Most of us joined FTC as unprepared rookies, **only realizing too late how much groundwork should've been done before the season began**. Because we lived that struggle, we now **hope use it to guide our outreach** and focus on helping students who are underserved or stepping into FTC without support, just like we once did. **This quote reinforced our mission to turn that experience into impact**, helping future teams start their journey with confidence and direction."

CONNECTING

with professionals

Georgia Tech



Goals:

Gain advice on how to manage a team, how to overcome software issues, how to motivate our team, learn a bit about FRC, and how to ensure that the robot is able to be very consistent

WHO:

Paarth Mishra, **FRC 113 ex-captain**, alumni, and **current Georgia Tech attendee**

TEAM TAKEAWAYS:

Ritvij - "Talking with Paarth gave me a better understanding of **how to create consistency in performance** through **smarter preparation**. He emphasized how tools like dashboards can help visualize problems and how small things such as **checking battery voltage or accounting for field differences**, can have a huge impact. It reminded me that **success doesn't always come from big changes, but from being intentional with the details** and learning from the teams who've already solved the problems we're facing."

Deekshita - "What I took away from the meeting with Paarth was **how important structure and routine are for performance**. Creating **detailed checklists, testing before and after matches**, and keeping the team organized by roles all help reduce uncertainty under pressure. His advice really drove home how much **FTC success depends not just on having talent, but on building habits that eliminate preventable mistakes**."

Isaac - "One thing that stood out to me was how Paarth talked about **learning from the top teams**. It's easy to feel like you're in your own bubble, trying to figure everything out yourself. But he made it clear that **connecting with others, asking questions, and watching how successful teams operate** is one of the smartest things you can do. That idea has pushed me to engage more with the wider FTC community and keep learning wherever I can."

CONNECTING

with professionals

Tyrex Group



Goals:

Learn how 3rd party manufacturers process orders, create environments of simulation for parts to be tested in, and how they have unique printers to create durable and uniquely shaped pieces



WHO:

Joel Coffman, VP of Marketing of Tyrex Group

TEAM TAKEAWAYS:

Ritvij - "Visiting the Tyrex facility was a valuable experience that highlighted the importance of testing. Mr. Coffman showed us various testing chambers used by companies to simulate real-world conditions like heat, water, cold, wind, saltwater corrosion, and rocky motion. **This for me emphasized how essential it is to test parts under stress.** For our season, even without high-end equipment, we can still apply this by **putting our prototype parts and mechanisms through harsh, realistic game conditions to ensure they hold up** during actual matches."

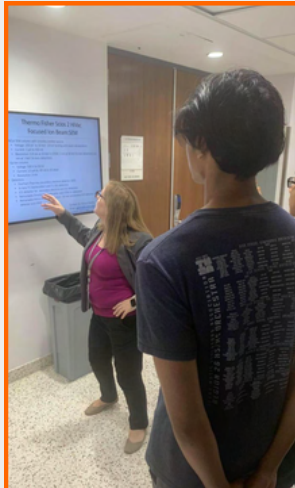
Isaac - "One part of this tour that stood out to me was the unique printers they had. Mr. Coffman explained how they have special printers to be able to print larger and stronger prints. This printer uses resin unlike the traditional filament. During this process, **the resin is cured and becomes solid.** This printing style allows for stronger design and allows for the creation of many parts at once. **I found the whole process greatly intriguing especially the part where they bathe the printed parts in alcohol.**"

Arjun - "My favorite part about this tour was the part where they showed us their **wire production.** Mr. Coffman said that the facility uses **2 million sq. feet of copper every year to make wires for clients.** We were then shown a live demo of **the wiring crimping process** which I found really interesting. Another part that I found interesting was the internship opportunities. Mr. Coffman explained how **interns are given free reign to a printer and computer and use their imaginations to come up with unique CAD pieces.**"

CONNECTING

with professionals

TMI



Goals:

Learn about the study of Material Science and learn about the different microscopes that are used by scientists to study different materials. Also, to learn about how to apply such material testing to robotics



WHO:

Dr. Ioana Gearba-Dolocan, **scientist and professor**

Dr. Andrei Dolocan, **scientist and professor**

TEAM TAKEAWAYS:

Ritvij - "Visiting the Texas Material Institute facility was such a great experience. One of the concepts that I learned from this experience was how they use **certain microscopes to scan surfaces of the materials**. Another concept that I learned from this tour was how scientists would use the transfer of electrons to see the quality of the materials. Dr. Andrei also explained how there were different **specialized microscopes depending on what was needed to be scanned**."

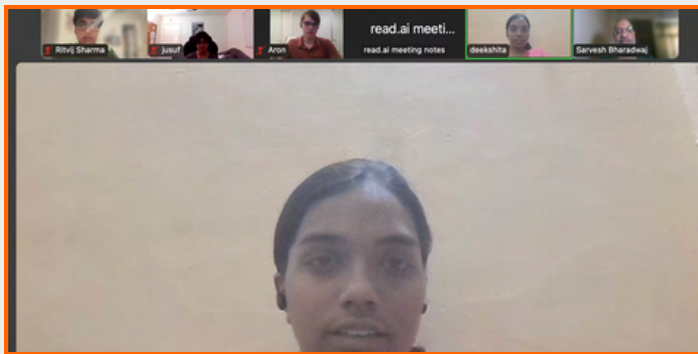
Jusuf - "During the Texas Materials Institute, Dr. Andrei and Dr. Ioana demonstrated how batteries work and took us into the lab to further explain the parts needed inside a battery. One concept I learned from this was **the transfer of electrons and charges between two parts of the battery in order to generate power**. I feel that this helped me **better understand how our own FTC batteries work and will help me know how to approach the debugging of them this season**."

Isaac - "One concept that Mr. Andrei explained to us was how **material testing centers are used by many major companies such as Tesla to test out their product's materials** before mass producing it. I felt this was greatly **similar to the Tyrex facilities but to a much more scientific extent** as unlike the Tyrex testing chambers, TMI was **able to view materials all the way to an atomic and chemical structure level** which could prove much more useful for some companies."

CONNECTING

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UKG



UKG

Goals:

Gain any potential advice from a Senior Software Engineer from the United Kronos group on leadership, crisis response as well as tips for passionate STEM students trying to make a career in Software or Engineering

WHO:

Mr. Sarvesh Bharadwaj, Sr Director of Engineering

TEAM TAKEAWAYS:

Aron - "It was an honor to speak with Mr. Bharadwaj. One concept that I remember well was when Mr. Bharadwaj was explained how to lead a large engineering team required setting clear, measurable goals, identifying risks early, and tracking progress consistently. For our FTC team, **this could translate into creating a structured build and competition plan, assigning responsibilities based on team strengths, and proactively managing potential issues.** Balancing technical innovation with **strong communication and teamwork is essential to running an efficient and successful robotics season.**"

Ritvij - "Speaking with Mr. Bharadwaj was such a great experience! In our meeting with him, he explained how having a strong technical foundation was essential to his success. This was shown when he explained how his expertise in mathematics allowed him to smoothly transition into becoming a software engineer. This deeply resonated with me as it showed me how **major of a role academics truly can play in coding especially mathematics**. Another strategy I learned from Mr. Bharadwaj was the importance that personal projects have on universities and companies. This also proved that **experiences such as FTC can lead to greater learning in software and can thus elevate one's resume with experience.**"

Deekshita - "One suggestion that Mr. Bharadwaj had for us was **how to face setbacks.** He said his advice was to **stay calm in the moment, attempt to resolve the situation as best as you can, and then after it is over, assess what went wrong and what needs to be changed to ensure it won't be an issue long term.** I feel that this could also be applied to FTC, in terms of prioritizing to fix the long term effects to ensure the problem won't arise again."

CONNECTING

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CNS Makerspace



Goals:

Learn about the machines used in the Makerspace. Learn about the different opportunities that they offer to high school students as well as learn about the projects created by other students.

WHO:

Ms. Rhiannan Morris, **UT**
Student and staff member

TEAM TAKEAWAYS:

Aron - "It was a great experience to visit the CNS Makerspace. One part of the tour that I found really interesting was some of their **projects**. Ms. Morris explained how they have a **Summer Program meant for High Schoolers** which gives them 5 weeks with access to printers, a glow forge, and a cnc machine and allows them to brainstorm ideas for a summer project to make. Some of the **projects that she showed us included a prosthetic arm made to help disabled students pick parts up. This arm was made out of servos and lines of pulleys which I felt was strung in a similar way to our own FTC Misumi and Viper slides.**"

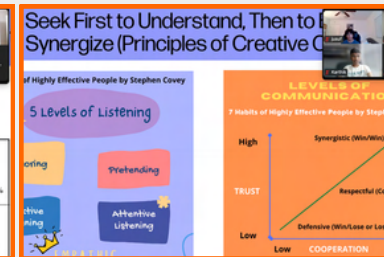
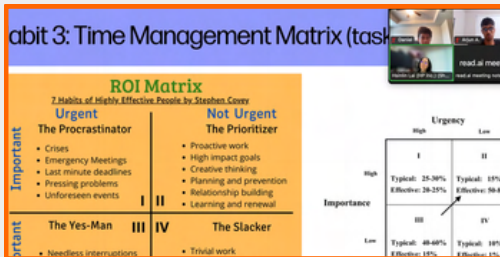
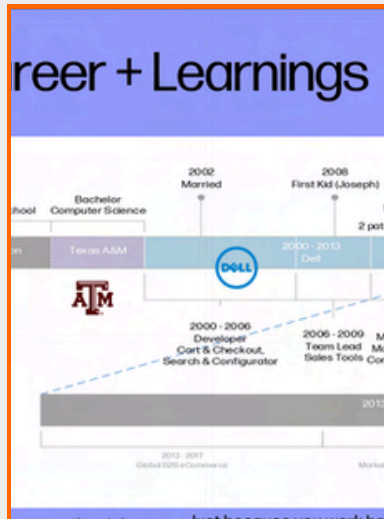
Ritvij - "Visiting the CNS makerspace was such a great opportunity! One takeaway that I had from this tour was the variety of equipment they had. **Ms. Morris had explained how they had special types of printer.** One which would **print out of Resin similar to the one we saw on the Tyrex tour** and then **one printer that would print out of metal dust to print stronger and more metallic structures.** Another part of the tour that I really enjoyed was how open the space was."

Jusuf - "Something that I really enjoyed learning from this tour was **seeing all the projects.** My favorite project was the **Iron Man mask which could detect for those who have high levels of anxiety their anxiety levels and then notify the user their need to stay calm in the situation.** I found their High School Summer Program greatly interesting due to its **application of microcomputers such as raspberry pi's and arduino's** to power such devices helpful to the community."

CONNECTING

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HP



Goals:

Learn from an IT Executive at HP the qualities needed to successfully function as a team, how to overcome adversity, the mentality to have for the season, how to motivate a team to continue to work hard, and how to maintain good relations with other teams.

WHO:

Mrs. Hsinlin Lai, IT Executive at HP

TEAM TAKEAWAYS:

Karthik - "I greatly enjoyed the teachings of Mrs. Lai. Her experiences and our discussion with her made me realize we have **a lot to learn in order to become an optimal team**. One piece of advice that Mrs. Lai gave to our team that really resonated with me was **how to motivate a team**. Mrs. Lai explained in her workspace, some ways she **motivates her team is by recognizing hard work, avoiding playing favorites, and clearly giving instructions that are specific and understandable to avoid confusion**. I feel that **this could be implemented into our team to be able to allow everyone to contribute equally which was an issue that I noticed from last year.**"

Ritvij - "Meeting with Mrs. Lai was an inspiring experience. One big takeaway for me was **how crucial time management, a growth mindset, and empathy are to building an effective team**. She shared how she uses tools like Agile to stay organized similar to Mrs. Srivastava's advice on setting priorities. I appreciated how she **emphasized always striving to improve and understanding others' situations**. I think we can **apply this by learning from and supporting other teams similar to how Paarth and now Mrs. Lai both encouraged**, regardless of past experiences."

Daniel - "It was a great opportunity to meet with Mrs. Lai! One of the most important thing I learned from Mrs. Lai was the **value of exploring diverse computer science pathways**, such as data science, cybersecurity, and systems engineering. She also recommended using tools like **JIRA for team communication and encouraged open source projects**. I think we could apply her strategies to FTC by **giving team members the opportunity to try different roles, promote system wide debugging, and use real workspace tools like Github to improve team collaboration**.

CONNECTING

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Texas Robotics Tour



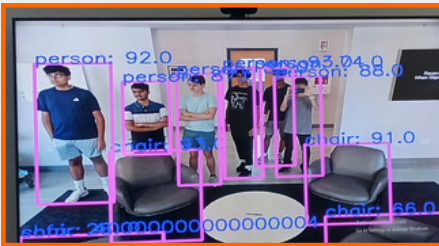
TEXAS Robotics

Goals:

Learn about the machines used in the Makerspace. Learn about the different opportunities that they offer to high school students as well as learn about the projects created by other students.

WHO:

Carman and Jonathan, Texas robotics guides and UT students



TEAM TAKEAWAYS:

Aron - "It was a fun experience to visit the Texas Robotics facilities at UT Austin. One takeaway that I had from this tour was the **applications of robotics in medicine**. On the tour, Jonathan, a **UT student** showed us a robot that could mimic precise surgical movements. In the lab, Jonathan demonstrated how the robot would perfectly execute the action he did and explained this was a **monumental milestone for medical surgeries that beforehand may have been too risky to conduct by hand**. Jonathan also explained that this also came with the **drawback of no tactile(touch) feedback and thus would be hard for the surgeon to feel whether they are being rough or smooth**"

Ritvij - "I really enjoyed visiting the Texas robotics facilities. One aspect that I really enjoyed was how they **used cameras in order for robots to identify objects**. In the first room we went into, we saw an application of tensorflow for object detection by repeatedly using hundreds of images to train the camera to identify objects. I found this really cool since in many past seasons this was used during FTC randomization to help score more points in autonomous."

Arjun - "One part of this tour I was fascinated by was the **number of robots that were able to conduct human like motions and be able to sense its surroundings and adjust its path**. One robot that we encountered was able to move around the **entire building autonomously** and was able to go around obstacles or stop to wait for the obstacles to move. This was similar to robots **I had seen other places which used such bots for food delivery at restaurants**. Another bot that we saw **was a humanoid bot that could mimic human like motions with great precision**."

CONNECTING

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Stanford



Stanford
University

Goals:

To learn about the academic, professional and personal journey of a Stanford alumni through various aspects like the technical skill, mindset, and long-term planning

WHO:

Anjana Rajendran, **Stanford Alumni, Electrical Engineering Major**



TEAM TAKEAWAYS:

Ritvij - "Talking to Mrs. Rajendran was an incredible experience! One takeaway I had from this meeting was the importance of fully planning out a project before starting on it. Mrs. Rajendran claimed that **"a project is only as good as its overall system design"**. I think one way we can apply **this principal as an FTC Team is to have lengthy discussions together for the first few days, in-person as recommended by Mrs. Rajendran to be able to better understand each other and together come up with a design structure** before starting their design process. "

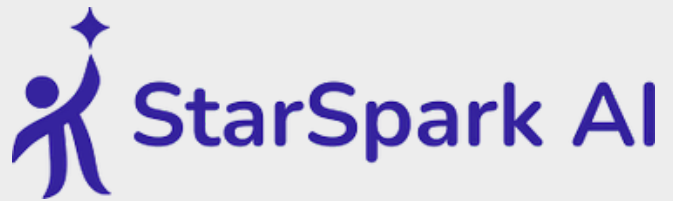
Aditya - "One takeaway I had from this meeting was how to overcome adversity and stress. Mrs. Rajendran had a unique strategy to overcome stress at her own office where she would work with a calm mind and would try and **break major daunting tasks into small manageable tasks and reward herself after completing each small task** I feel we could apply this into our robot assembly and programming by **not looking at the end goal but the much smaller steps to take to reach that to make it feel much more doable.** "

Arjun - "Mrs. Rajendran told us this meeting about the importance of **good collaboration in a team**. She explained that many times **when a team first meets, they have trouble working together**. But, **through mutual experiences, they learn how to work together better**. I feel that **we are applying this by having everyone regardless of department participate in outreach and tours so that they can interact with each other and learn more about each other**. Another principal that Mrs. Rajendran told us about was the **"Pareto Principle, which plagues many teams where 80% of the work is getting done by 20% of the team"**. Mrs. Rajendran suggested that everyone have a general idea of everything so that anyone can substitute anywhere if an issue arises. "

CONNECTING

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Starspark AI

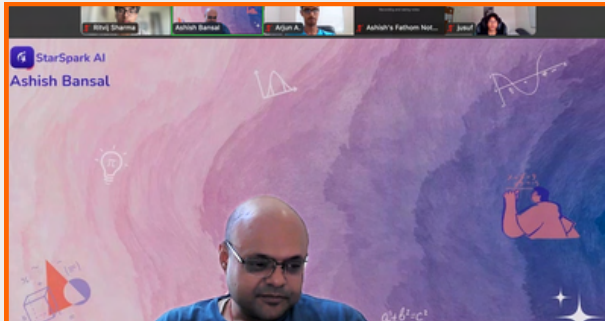


Goals:

To learn about the impact of AI in the STEM industry today, and potentially how it can be used as a tool in today's FTC in processes such as Autonomous and strategic scouting.

WHO:

Ashish Bansal, **Starspark AI**
CEO



TEAM TAKEAWAYS:

Ritvij - "Talking to Mr. Bansal helped us learn more about AI and its capacities. Mr. Bansal explained how many times **other Chatbot models are unable to provide in depth solutions or solutions** which challenge the user to arrive to the conclusion with minimal help to ensure they grow. Thus, he explained he **created StarsparkAI to be able to teach like a traditional tutor for homework help but from anywhere in the world** as long as you had internet connection. Another aspect that he explained that I found unique was that he had compiled multiple trained models together to answer questions which was unique from other chatbots such as ChatGPT and Grammarly."

Jusuf - "Speaking with Mr. Bansal was a great opportunity. **One question I had was how in many major companies, there had recently been data leaks which had led to people's personal information being leaked.** Thus, we asked Mr. Bansal how he managed that. He explained that in his company, during sign up, **the only requirement was the person's first name and account email.** Thus, there was very little actual data that would be leaked even if someone would hack into the database."

Arjun - "Mr. Bansal gave us some great insight about AI in this meeting. We asked him for his vision for AI as it expanded and he explained that in his view, **AI would control everything and would have a much bigger impact in robotics such as in household AI helpers that could make real time decisions and do many tasks for humans in the future.** Another concept that Mr. Bansal taught us was the **importance of concepts building on each other during their testing phase highlighting how they would start by feeding the models basic info and slowly build up in fast iterations and fast feedback from users** to optimize it."

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Tenstorrent



tenstorrent

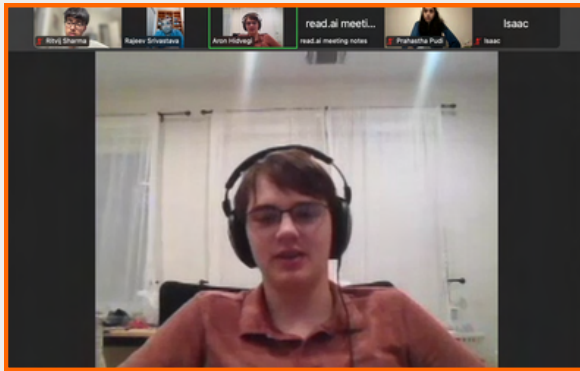


Goals:

To learn about the responsibilities of an Electrical Engineer at Tenstorrent, the current state of the Semiconductor industry, and any advice for High Schoolers who wish to study Electrical Engineering in the future

WHO:

Rajeev Srivastava, **Director of Physical Design at Tenstorrent**



TEAM TAKEAWAYS:

Ritvij - "Speaking with Mr. Srivastava greatly helped my understanding of the Electrical Engineering field. Mr. Srivastava explained how he was part of **back-end development of electrical chips**. He then explained how in his job, work is split into front end development which is the coding of the chip and then the back end development is meant to **translate the code for the chip written in a physical design using CAD tools** and to overall ensure that the chip can function efficiently and effectively. I felt that **this division was similar to the split of hardware and software which indicated to me that such teams that must work together exist in the industry** as well."

Isaac - "One learning I had from this meeting was the **importance of modular code**. Mr. Srivastava explained that as **a former FTC and FLL coach**, his one piece of advice he had for us was to start by programming **basic functions such as maybe extension or the actual drivetrain before fully optimizing it and putting it all together**. Another piece of advice he had for our team was to **make our autonomous not so battery dependent as he explained that in his experiences, battery dependent autos were much more likely to fail**."

Aron - "Meeting with Mr. Srivastava was a great experience. One concept that Mr. Srivastava told us about was the **difference in chips across different fields of STEM from computer chips to medical chips**. Mr. Srivastava explained the reason for this **was the need for power in some chips such as medical chips versus speed and efficiency in chips for electrical components**. From this, I feel that we implement this idea into our FTC season by having a **different type of bot depending on the season's game similar to how different chips are used for different reasons**."

CONNECTING

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Microsoft tour



Goals:

Visit the Microsoft office building and learn about the different teams and how they work together to work efficiently. Also to learn industry standard working processes to potentially add it's aspects into our process

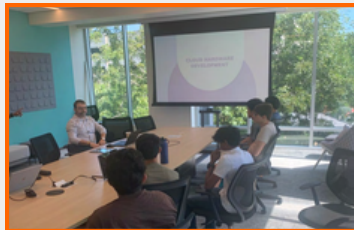
WHO:

Mr. Prakash, **Product manager**

Mr. James, **Program manager**

Mr. Narayanan: **Silicon Design Engineer**

Mr. Praveen: **Silicon Design Engineer**



TEAM TAKEAWAYS:

Ritvij - "It was a great time to go tour the Microsoft office. Mr. Prakash, Mr. James. and Mr. Ravi first took us through the office and then led us to a conference room and asked us what we wanted to know about their profession. We then explained how we hoped to learn the processes that Microsoft uses and the different products they have. **Mr. Prakash** started off by **explaining** how his **job as a product manager requires him to take the constant feedback from customers on how the product is working, sales reports** from Microsoft, **and report it to the Engineers to create constrains** for the design and programming team **similar to the constrains of the FTC game.**"

Aditya - "During **the presentation of Mr. James**, he mentioned how a microcomputer works requiring **billions of microprocessors**. He then explained how they all worked as a team with an example saying that **when Mr. Prakash reported that the market would need to prioritize speed on their chips over power**, they switched and had great resulting sales. **This showed me how every branch working together well allows for overall team success.** "

Aron - "One aspect of the presentation I enjoyed was how Mr. James compared working in the team was like a computer. He explained how a computer has many **different components to ensure it runs smoothly with a mother board which would control all the other aspects. He emphasized how important this was to be applied into teams.** Another learning I had from this meeting was the development process used by Microsoft called NUDDs(**New, Unique, Different, and Difficult**). This process was meant to analyze all the constaints and be able to **make an optimal design based on what was new, unique, different, and difficult.** I feel we could apply this into our team by **having some of these components of analyzing the constrains and applying it into the design in our own design process.** "

CONNECTING

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Inventionworks tour



The University of Texas at Austin
Texas Inventionworks
Cockrell School of Engineering



Goals:

Visit the largest UT Makerspace on campus and learn about the different machines that are used by the students and many of the projects that they are used for as well as the opportunities they offer for High Schoolers

WHO:

Benjamin Black,
Inventionworks staff and manager



TEAM TAKEAWAYS:

Isaac - "Visiting the Texas Inventionworks facility was a great experience. One part of the tour that I really enjoyed was viewing all the projects they had. One of the projects that I saw was a **real life working flamethrower made by a previous student for his senior project!** Another project that I saw was a **prosthetic similar too the one we saw at Texas robotics but this team driven my a sprocket mechanism** which was another graduated senior's final project."

Jusuf - "One part that I found really interesting about this tour was the **vast number of machines they had**. One machine I found specifically interesting **was the laser cutting machines**. Unlike our makerspace's glow forge, this machine had the ability to cut different layers and create texture on the wood unlike the glow forge in our makerspace. Another machine I found interesting was the Resin printer as unlike the other 2 tours we saw the Resin printer, this one we could see live to truly understand how it works."

Aron - "This tour was amazing. I got to see the live usage of many of the machines that we use in FTC in other places. One of the techniques I learned **from this tour was drag soldering which is meant to solder extremely small wire to extremely small ports on a motherboard**. Our tour guide and Mr. Black explained that many **students use "drag soldering" in order to make secure connections at a smaller scale**. We also saw the use of normal soldering used in their projects as well. However, in **projects with smaller components**, much more drag soldering was implemented."



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Gigafactory Tour

TESLA



Goals:

To Visit the Tesla Austin Factory and learn about the manufacturing process that Tesla uses to optimize their output. Learn how the factory is split up and the machines used in the assembly line

WHO:

Mr. Matthew,
Gigafactory tour guide,
Mr. Yerramreddy,
Software engineer



TEAM TAKEAWAYS:

Prahastha - "Visiting the Tesla Gigafactory was a surreal experience. **One takeaway I had from this was how the Tesla Battery was made out of Nickel, Cathode's and Anode's** (positive and negative charges) that create energy when they react together. **I also noticed that they were well organized with whiteboards with clear goals written in many subsections of the factory similar to the subsections of our team."**

Daniel - "Touring the Tesla Gigafactory in Austin was an incredible experience, especially seeing Tesla vehicles getting assembled right before our eyes. One thing that really stood out was its approach to integration and manufacturing process. **The Model Y assembly line uses horizontal integration (made left to right)** while the **Cybertruck uses vertical integration, like an upside down pyramid starting on the 4th floor and ending at the ground level.** We also learned that almost **everything is created in the factory, from iron casting the underbelly of the cybertruck to painting and even having the finished car drive itself to the parking lot."**

Aron - "Seeing the **manufacturing process at the Tesla factory** was a great experience. One takeaway I had was the implementation of HEVs that would follow a magnetic strip to speed up the assembly line. I also saw how robots such as **CoPa's were used to accelerate production by taking parts from the higher floors and lowering them down.** This usage of robots to automate production also showed me the application of robotics in our daily lives. Finally, I learned about **Tesla's customer service process called SPARC** to beat other competitors in the market. "

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Indeed Building Tour

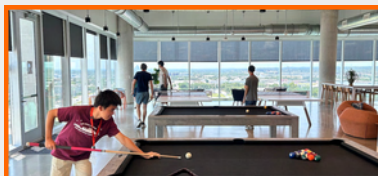


Goals:

To learn about key industry practices used to optimize workflow. Also to learn how Indeed uses in depth analysis to gain an advantage over their opponents, as well as the debugging process used when errors come up.

WHO:

Mr. Tim Kao, **Software Engineer and Senior Manager at Indeed**



TEAM TAKEAWAYS:

Karthik - "This was my first tour, and I really enjoyed it. Mr. Kao showed us around and explained both his background and how **Indeed uses AI to improve job matching and profits. When we asked about debugging, he said it's important to classify issues by severity and address them in that order.** This is a method I think our team could **adopt to stay more organized and efficient in our planning and our own debugging and bot repairs.**"

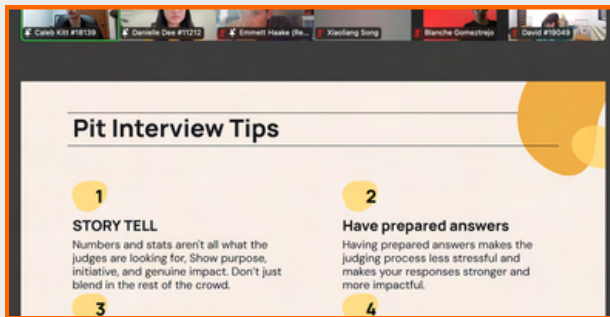
Daniel - "I really enjoyed visiting the Indeed office. One key takeaway was **Mr. Kao's point on simplicity in design.** I think that we can **apply this by having our full team carefully decide whether each feature is truly necessary.** He also emphasized the value of having everyone contribute across sub-teams, even if they're not familiar with a field. **Explaining and listening to each other can spark new ideas (known as swarming at Indeed).** Hearing this now from **both Microsoft and Indeed, I feel we should adopt it into our own team as it seems to be an industry standard.** Another process that I learned about was **RAID(Risks, Assumptions, Issues, and Dependencies) Analysis that Mr. Kao said many Indeed managers use to stay organized in their planning.**"

Aron - "Coming to the Indeed office was a great opportunity. I really enjoyed visiting all the floors and seeing how Indeed keeps their staff motivated and the different facilities they have for them to use. One of the takeaways I had from our tour with Mr. Kao was the **importance of analysis: The ability to understand your own strengths and weaknesses, your opponents strengths and weaknesses, and the resources you have available** and making the best possible based off that. **After hearing the same advice from both our student mentor Peter and Mr. Kao, I think this reinforces the need for us to implement this into our own analysis of the game and building our robot.**"

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Inspire Summit



Goals:

To Learn the strategies from Multiple Time Inspire Winners that help them win the Inspire Award for multiple years. Also, to ask them questions of their secrets to getting many sponsors and making long term impacts in their community.

WHO:

Emmett Haake, Rebel Robotics portfolio designer
Danielle Dee, The Clueless outreach captain

TEAM TAKEAWAYS:

Ritvij - "It was great to attend the Inspire Summit hosted by Rebel Robotics and The Clueless. One takeaway I had from this meeting was the **effectiveness of cold emailing companies**. However, Emmett explained that many times companies do not want to just read emails inquiring about funds and that instead **we should aim to request to potentially meet so you can show them your passion which could incline them to sponsor you.**"

Deekshita - "One piece of advice we received from the summit was to **have supplementary materials for everything. They explained that it would be easier to explain our story if we would have pictures, documents, and old physical iterations of our robots at our pits and in our presentation.** Danielle also explained the importance of having outreaches that would have a long term impact and would go beyond just that day. I think this is a very good idea to implement into our future outreaches. Another cool idea **I learned was to have 1-2 leads per outreach to be able to execute more and have the full team involved in it.**"

Jusuf - "One takeaway I had from this meeting was the importance to make **portfolios very readable and skippable.** One specific piece of advice I remember from Danielle was to make your port **pages so the judges do not require context** to be able to read each page and **make parts that you want the judges to read flashier and more eye-catching.** Another piece of advice that I remember from Emmett was the importance to **remember in your head what points have been covered as part of the criteria of the award you are aiming for** during both pits and formal judging. **Emmett also explained that in pits AND formal, the judges time is limited and we should strive to hit as many points as we can even if it means cutting off our teammates to hit our points.**"

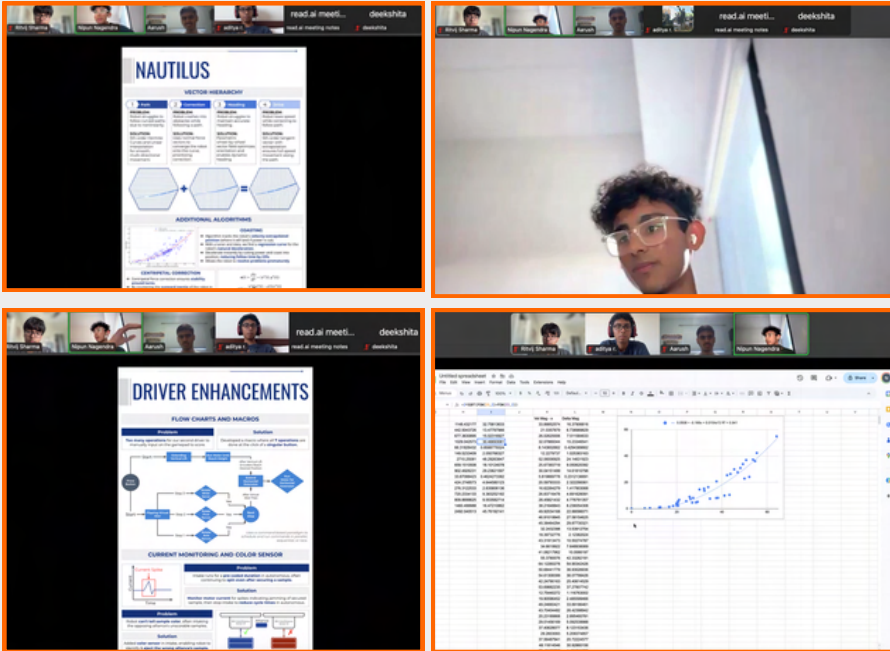
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Nipun Meeting



Cornell University



Goals:

To learn tactics to optimize our presentation and judging overall. Also to learn how to optimize both TeleOp and Autonomous by using advanced math and trajectories. Also, to learn how to clearly explain the software side in judging clearly.

WHO:

Nipun Nagendra, **Cornell Student and former 6210 Stryke captain.**

TEAM TAKEAWAYS:

Deekshita - "It was really interesting to meet with Nipun. Nipun started off by explaining how **he created his own custom path follower last season** which was something interesting as I had never seen teams make their own path follower. Nipun explained the benefits of it saying it **made it easier for him to write code** since he was the most familiar with it but also explained the **downsides being that it took multiple months to fully debug and that there was limited help available to debug it.**"

Aditya - "Meeting with Nipun was a great experience. He started off by explaining how to truly talk about software to be able to convey its effectiveness to the judges. **He explained how we should explain issues we faced in software and how we were able to overcome them in the code.** Another concept that Nipun explained to us was **how to make a guided vector follower, Pedro reactive follower, and friction compensation methods by testing values in our teleOP to make it smoother overall.** Nipun also explained how his team **tuned both the drivetrain and slides by adding feed-forward controls that cancel out natural forces like friction and gravity.** This made the robot move more smoothly, so the driver's inputs directly controlled its motion with greater precision."

Minessh - "Nipun gave us some really good advice in this meeting. One piece that definitely resonated with me was his advice on using **tools such as ChatGPT and Github Copilot to help optimize your code** to be able to perform faster and better. Another piece of advice from Nipun was to have some **software principals** to follow. He explained that for his team they followed **KAIZEN**, a phrase meaning **"change for the better"** in Japanese. Nipun explained that his team used this phrase **to help identify weaknesses through constant testing and performance reviews.**"



CONNECTING

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Abraham Meeting



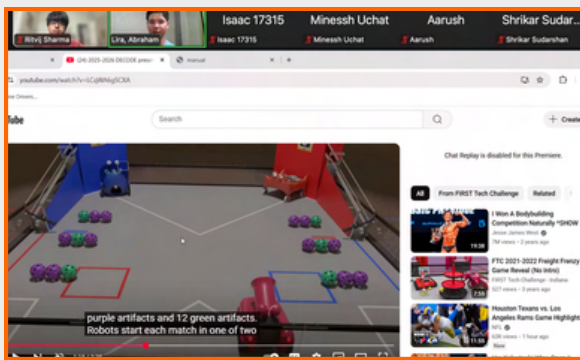
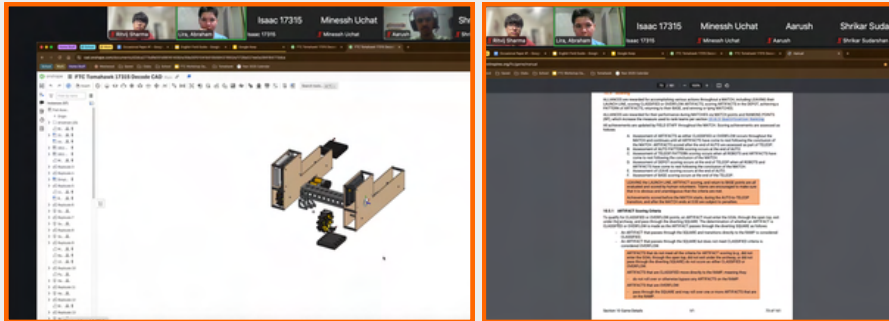
TEXAS A&M UNIVERSITY
Engineering

Goals:

Learn from a former FTC Edison Division winning alumni how to have sustained success, how to create a winning formula to have a good robot, and how to network with other teams to improve your chances for alliance selections.

WHO:

Abraham Lira, **former 2023 Tomahawk alumni and current A&M Engineering student**



TEAM TAKEAWAYS:

Shrikar - "It was a great experience to meet with Abraham. Meeting with him gave me a really good amount of insight for the upcoming FTC season. One piece of advice that he gave us was to prioritize simplicity and consistency over more complicated mechs explaining that although certain mechanisms such as a turrent and a swerve drive seem ideal at first glance, they may have drawbacks. Abraham specifically advised us to not use a swerve drivetrain this upcoming season since we didn't have much practice for hardware creating it and then software programming it. Abraham specifically explained that when his team made worlds in 2023, he met a team that had tuned their swerve drivetrain the entire season and summer and yet at worlds struggled with consistency."

Aditya - "It was an enlightening experience to meet with Abraham. In our meeting, we talked a lot about how to create a worlds-worthy bot from the software side. He emphasized the critical aspect that we should limit the usage of ground-breaking technology or unstable software as it is not a tried-and-tested solution and as a result can mess up results when it matters."

Ritvij - "Meeting with Abraham was a great opportunity to learn from more experienced mentors. Abraham explained to us the importance of creating connections this season for potential alliances. I then asked him for his advice on how to do so and he replied to simply walk around competition and talk to other teams as the better people knew you and your team, the higher chances that you would be selected and could make a case to be selected even if you didn't have the best record."

CONNECTING

with professionals

Zaiden Meeting



Goals:

Learn from the experiences of people who were successful in FTC, and learn from them what they did in order to be able to succeed so much and how they were able to overcome adversity in tough times.

WHO:

Zaiden Dee, 11212 The Clueless alumni, former captain, and FTC Centerstage World Champion

TEAM TAKEAWAYS:

Sanaika - "Zaiden explained how their team balanced technical work with outreach, which helped them win awards and build a strong reputation. **Outreach projects were planned in the off-season**, with each member owning a project under the guidance of an outreach lead. This was similar to what we had done already as the planning for many of our projects were started early in the offseason. **Regular check-ins kept everyone accountable**. Zaiden also highlighted the need for clear communication between hardware and software teams, recommending frequent updates to keep the team aligned and avoid friction. "

Deekshita - "Zaiden gave us some really great advice! Zaiden explained how important robot simplicity is. his biggest piece of advice from him was to **make a simple robot** and instead of rebuilding like many other teams suggested, he advised to instead **continue to build upon what we already have**. He said complete rebuilds could be risky and inconsistent and instead suggested focusing on a consistent auton and TeleOp when possible. "

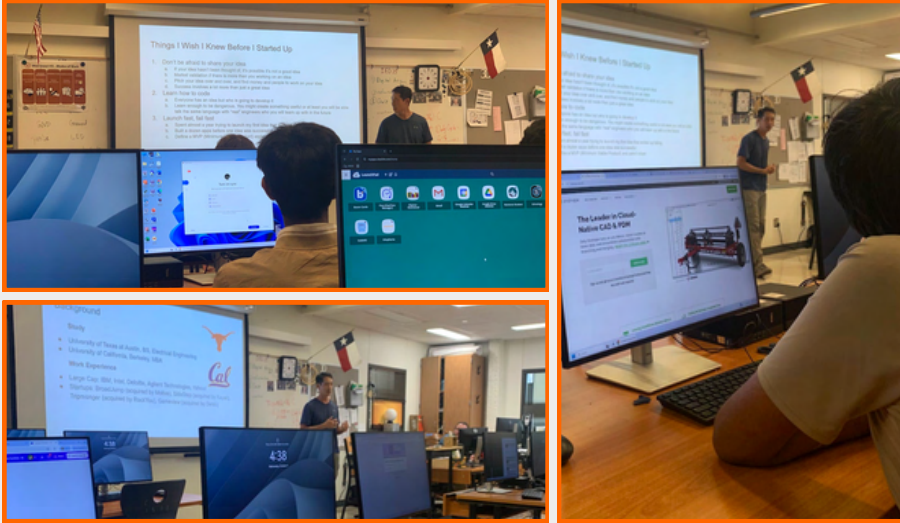
Ritvij - "Meeting with Zaiden Dee was a great experience. Zaiden started off by explaining how in his view, it was important especially for alliance selection to have **constant communication with other teams** throughout the event so they **remember who you are**. Similar to Abraham's advice, he said that if you're in a situation in where you aren't the highest rank because of unlucky qualifications, **you need to be strategic in what teams you decide to pitch a potential alliance to**. This is because between the alliance selection rounds and the actual qualifications, there is very limited time. Thus, Zaiden **advised us to speak to lower seeded teams to increase our chances to get picked in that moment compared to approaching higher seeded teams** unless you have a strong connection with them "



CONNECTING

with professionals

John Hwang Meeting SideStep



Goals:

Learn from other industry leading professionals on how to create industry leading applications and the pieces of advice they may have for us on how they succeeded.

WHO:

John Hwang, **Founder of BroadJump and SideStep**

TEAM TAKEAWAYS:

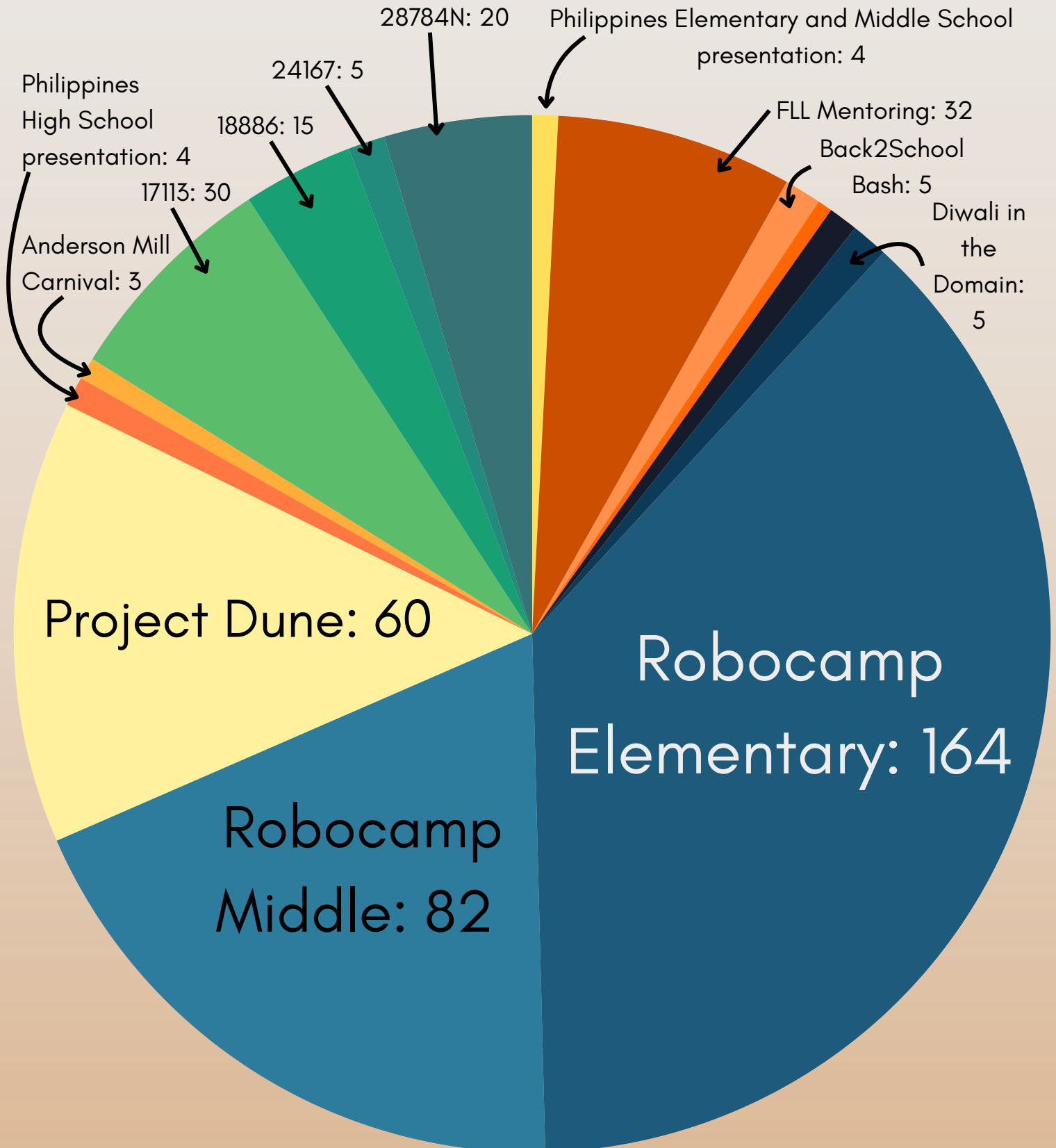
Ritvij - "Meeting with Mr. Hwang was a great experience. Mr. Hwang started by explaining a bit of background about himself such as him having an **Electrical Engineering undergraduate degree from the University of Texas at Austin** and having an **MBA from the University of California at Berkeley**. Some piece of advice he gave was to not be afraid to share your idea as communication with others can help refine your idea and make it better. This was a unique piece of advice that I had heard since many times before, **I have heard that it is important to keep your ideas secret but this gave me a new perspective** on how to approach collaborating with other teams. Another piece of advice that Mr. Hwang said was to use the **KISS principles (Keep it Sweet and Simple)**. I felt that this was a great idea as it was **similar to what we had previously learned from our other mentor Mr. Uchat**. Another piece of advice that Mr. Hwang gave was to **develop on existing API's** when developing a new product. Although not directly applicable to our season, I feel that we could still **implement this advice by building upon our work from past years of FTC** especially in terms of a network of mentors."

Arjun - "Meeting with Mr. Hwang was a new experience altogether due to the unique perspective he offered regarding business in tech. As a professional who has **experience with Engineering in tech giants such as Dell, and with several startups such as SideStep**, he offered valuable insight into ways young coders and engineers can develop their ideas and bring them to life. Through his unique perspective, I was able to learn how to approach collaboration within a team and contribute ideas in the most appropriate manner. Additionally, I felt like his insight also matched some of my thought process when he emphasized KISS and explained the results and impacts of simplicity, **encouraging me to consider more efficient and simple alternatives to bot related mechanics**. Overall, I believe this experience was unique altogether due to the fact that unlike previous professionals we spoke to, **Mr. Hwang had both business and tech experience, which allowed us learn about how to grow and spread projects**.

TEAM REACH

Total hours we spent impacting our community and beyond

HOURS SPENT

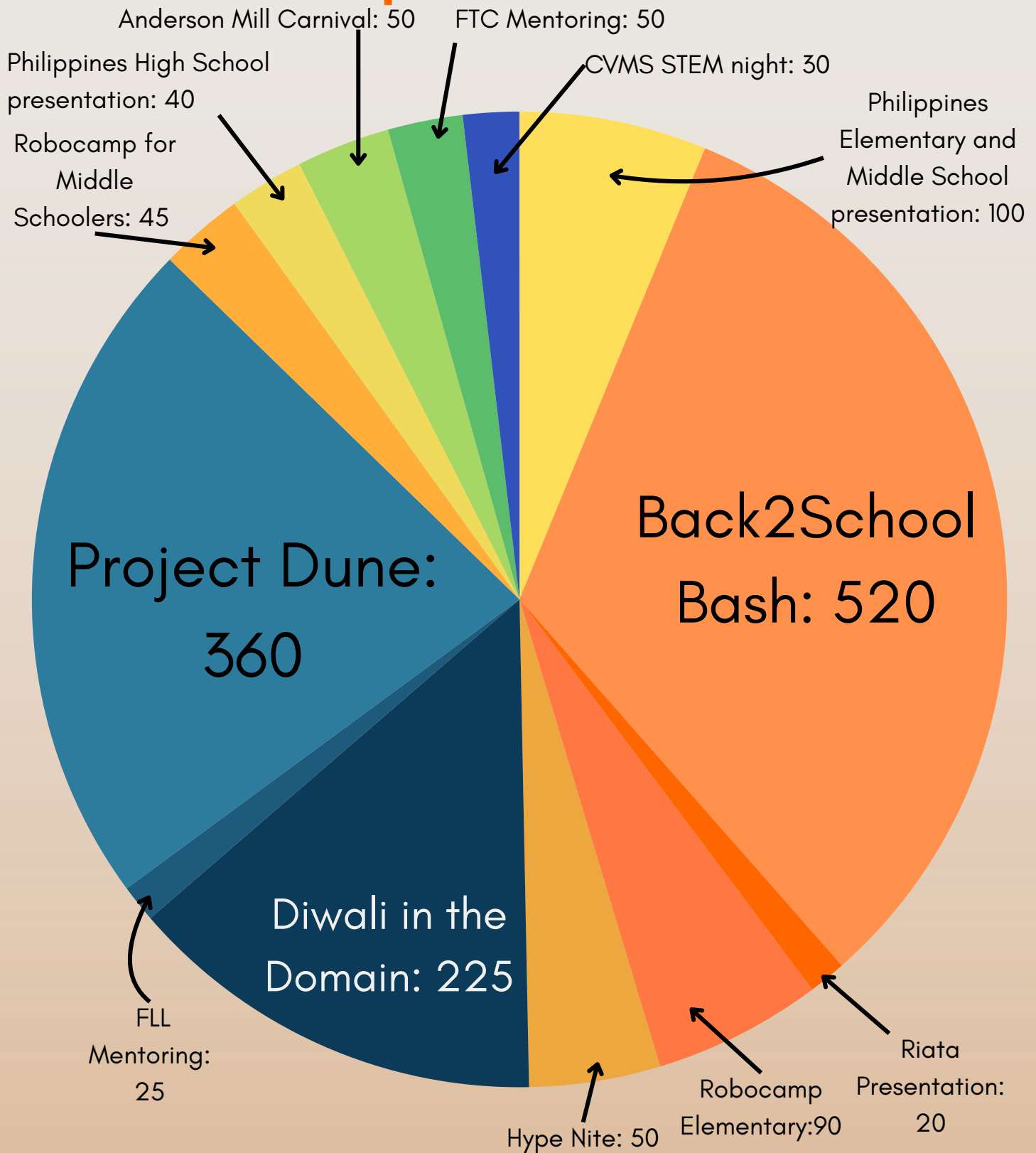




TEAM REACH

The number of people we were able to reach with our efforts

People IMPACTED





Motivating

our community

Philippines Outreach- Elementary & Middle

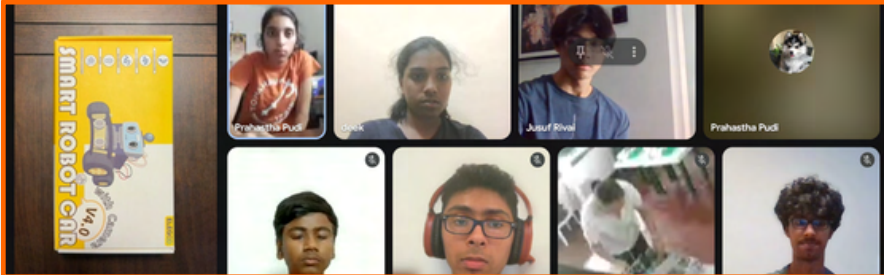


GOALS:

Introducing robotics to elementary, and middle school students through giving kits and guiding their assembly

WHO:

Our Lady of Joy Learning Center, a K-12 learning center located in Consolacion, Cebu, Philippines



TEAM TAKEAWAYS:

Arjun - "Working with the students at Our Lady of Joy Learning Center was truly enriching. I introduced them to the **Elegoo Smart Car that came with motors, sensors, wires, and microcontrollers**. I was truly amazed by their curiosity and engagement throughout. **They asked thoughtful questions and followed each part of the presentation closely**. While building the bot, **one student even caught a small mistake I made** with the motor orientation, which really impressed me. Despite limited resources, **their eagerness to learn and quick understanding showed me that, with the right support, anyone has the potential to become future innovators.**"

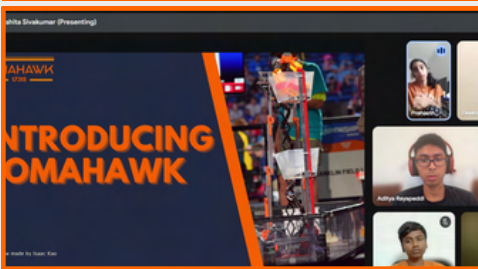
Jusuf - "My favorite part was **seeing the students' excitement turn into rapid progress right in front of me**. After a short walkthrough of motors, gears, and batteries, I built alongside them on camera, offering quick pointers whenever they needed help with the building process. **They kept pace effortlessly which showed just how eager they were to create their robots**. **Knowing that opportunities like this are still uncommon for many schools in the Philippines made their teamwork even more inspiring.**"

Karthik - "This was one of my favorite outreaches. My job was to teach the children in part about the **Hardware aspect of FTC as well as assist the students in constructing the kits**. One of the quotes from the school that really **resonated with me was when the principal said "Thank you very very much to you and your enthusiastic team!!! Yesterday was truly bright and sunny as you all brought new experiences, new memories and new hope for all that the future holds."** Before this, I had participated in our outreach impacts but **after seeing the principal's gratitude and seeing the smiles on the children's faces, I want to help out in more of these events.**

Motivating

our community

Philippines Outreach - High School



GOALS:

Use kits to introduce simple robotics, as well as introduce FTC to the High Schoolers and teach them the basics of how the robots work and how the competition works.

WHO:

Our Lady of Joy Learning Center, a K-12 learning center located in the Philippines

TEAM TAKEAWAYS:

Aditya - "During the Philippines Outreach program, **I taught robotics software using Arduinos and ESP32 chips to fellow high school students.** It was a meaningful experience where not only did I learn something new, but I also got to share it globally. It **showed me how FTC empowers us to make a real impact through outreach, which has become a deeply personal part of my robotics journey.** The support from Mr. Creer and the school principal reminded me that lasting change starts with taking initiative."

Deekshita - "My favorite part of the experience was during our highschool section. Half the time of this section was spent explaining FTC to the kids. **They were very curious and asked many questions.** There was definitely one part way more memorable than the rest though. After our introduction, Prahastha asked the kids how they felt about **potentially starting the first FTC team in the Philippines, and they erupted in cheers. They were extremely enthusiastic and I look forward to working with them in the future!**"

Prahastha - "Working with high school students in the Philippines was an incredible experience. **I had the opportunity to introduce them to Spiderbot, and it was so much fun building and exploring robotics together.** They were **genuinely curious and enthusiastic, which made explaining different hardware components** and their functions even more rewarding. Their excitement, eagerness to learn, and **willingness to dive into hands-on work made the entire session feel energetic and inspiring.** It was a wonderful day of learning, connection, and shared passion for robotics."

Motivating

our community

Back to School Bash



ROUND ROCK
INDEPENDENT SCHOOL DISTRICT

GOALS:

Use our robot to motivate Round Rock ISD youth and other attendees to have a greater interest in robotics and also connect with high level members of the RRISD board to discuss the need for more equipment for RRISD robotics programs.

WHO:

Back to School Bash for **students of Round Rock ISD, other event attendees**

TEAM TAKEAWAYS:

Ritvij - "This was one of my favorite outreach activities. **There were almost 6,000 people who attended** which was very exciting. What we would first explain the controls of the robot and give them a small challenge which was to push 4-5 samples into a blue tile in under a minute. **Watching the kids smile's after scoring a sample, even if it was only one made me very proud.** This event also showed me the curiosity and excitement that robotics brings as **the line to drive for a minute with the robot was almost always at least 10 people long!**"

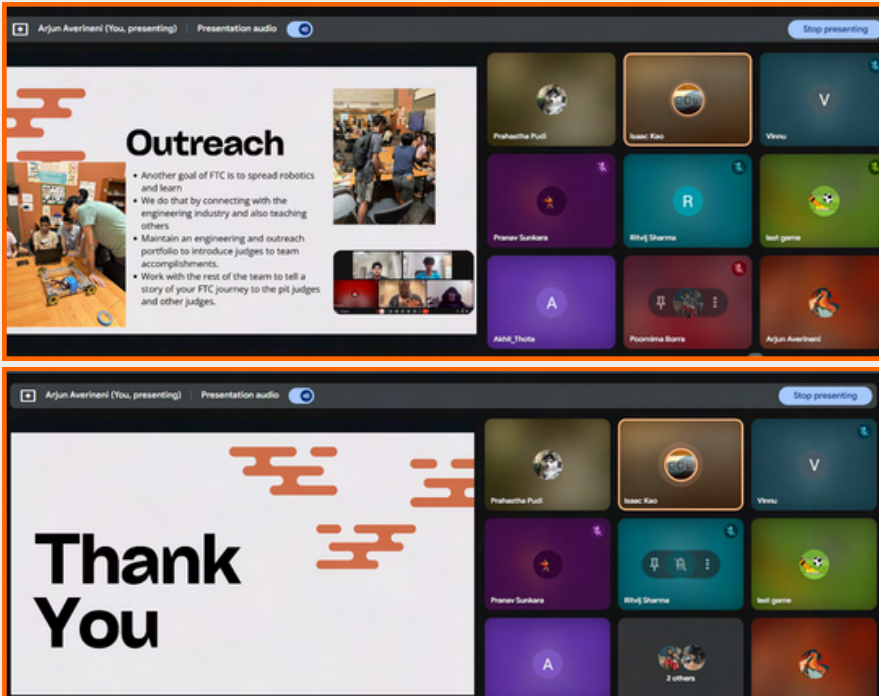
Arjun - "This was a gret opportunity in my opinion to reach out to many people due to kids from many different schools coming. One of my favorite memories from this outreach was **meeting and discussing with the RRISD Superintendent and many other RRISD District Board members.** Meeting with them revealed to us that **they themselves were interested in our robot's design and our process.** They then explained how they were excited to see our commitment and **agreed to consider a potential expansion for our school program as well as bringing up potentially increasing the budget for STEM activities at the next board meeting."**

Jusuf - "Coming to this event was incredibly fun. One of my takeaways was **the curiosity of the children.** One of my favorite memories was the amazement on the faces of the children when they **saw the robot strafe** and many began to **ask how it works.** Then we explained how mechanum wheels create holonomic movement and how the structure of the wheels with the **rollers being at a 45 degree angle and the rotation of the wheels in a certain way would allow for movement to go in any way** including side to side and diagonally without turning."



Motivating Riata Community Presentation

our community



Goals:

To meet with students from our community and explain to them the process of FTC to raise awareness for FIRST Robotics and to potentially motivate them to start their own team in the future, give them resources to learn from if they want to pursue FTC

WHO:

Riata community middle and High School residents.

TEAM TAKEAWAYS:

Arjun - "I really enjoyed this meeting. In this meeting, we started off by explaining what FTC is and the benefits of doing it in High School. We then explained how there are **many different sub teams** involved in making the robot some of the sub teams that we explained existed included the CAD team, the Hardware team who would build the robot, the software team that would be responsible to code the robot. we then showed them a few videos of our robot to help spark their interest."

Isaac - "It was a great experience clearing up all the questions that the kids had from the presentation. Many of them were **curious about the advancement system in Texas** and thus, we explained how the advancement works. One of them asked us a question over **whether it was easier as a rookie team to advance from awards or robot. We then explained that in order to win from awards, it is vital that your understand the rubric completely** and then apply it into your team . So, many teams only win awards in their 2nd ,3rd , or 4th year and beyond. "

Prahastha - "It was a great experience to teach the kids how FTC works. After the meeting ended a few of them asked to stay afterwards to **ask about how to start a team.** We explained how in order to start a new team, they would **each need to pay ~200 dollars for their 8 people.** We then explained that there are **many grants for teams to get more funding. We explained that applying for these grants** would be a much better method than to pay high dues. "

Motivating

Robocamp

Elementary School

our community



Goals:

To teach elementary schoolers learn about how to code in basic blocks, have them learn the science behind basic experiments, learn how to make well working FLL Bots and learn how to code their bots using blocks.

WHO:

Elementary Schoolers from all around **the greater Austin Area**

TEAM TAKEAWAYS:

Arjun - "This was such a great experience! Robocamp was a camp that we and our organization ran together to help inspire kids to get into STEM related activities. Some of the activities we had **included activities such as making lava lamps, pop rockets, and extracting the DNA of Strawberries.** Overall, I really enjoyed helping the kids learn the science behind these experiments. When they saw how pop rockets worked, they were awed and asked how it worked. **It was so much fun to explain that the Alkaline-Seltzer reacts with water to create the explosion they saw!**"

Prahastha - "One activity we had the kids do **was the hour of code** where they would learn basic block programming. After their hour of code, we **showed them the maze that they had to program for.** At first many **kids were uninterested in programming, but then after seeing other kids running to the table to measure the distance and coming back telling us values, they asked to program as well.**"

Ritvij - "Teaching the kids how to build FLL battlebots was such a great experience. At first many of the kids were reluctant to build the bot but as we guided them and sparked their interest using an objective(a maze) to complete, many of them began to show interest and began to assemble their robot together. One of the **issues I remember the kids had was being able to attach the motors securely to the brain of the robot and so a unique solution they came up with was to try and use some of the blocks to lower the motors to make more space for it to fit and would just secure the block to the brain.** "

Motivating

our community

Robocamp Middle School



Goals:

To teach Middle School children how to create an FTC level Chassis and be able to write basic code to program the robot. Also, to demonstrate our competitive FTC bots to them to try and inspire them to join robotics.

WHO:

Middle schoolers from the **greater Austin area.**

TEAM TAKEAWAYS:

Arjun - "As part of the Middle School week, we attempted to have the kids learn how to make a basic FTC straffer chassis. At the start the kids were confused and asked for help. But after we helped guide them into screwing the motors on and explaining how the bevel gears worked on one side, **the kids were able to assemble to rest of the drivetrain with very little counselor help.** One question that they asked for me when it came time to test the drivetrain was **how it was able to move side to side.** Thus I explained how the **45 degree angles on the rollers of the wheels and the diagonal direction movements being opposite allowed for such movement.**"

Ritvij - "Teaching middle schoolers about FTC was a great experience. This camp was a lot more hands on. **First in the week, we had them all register for Onshape and learn the basics. The first part we had them make was a simple rectangle that had a hole inside. Then, we attempted to teach them how to create a basic claw to pick up game pieces in FTC.** I was delighted to see how some of the kids really enjoyed creating the claw."

Deekshita - "After we had the kids design a chassis and have them make a claw, we decided to teach them FTC program. First, we showed them **how to configure all the motors in the code to identify each motor such as frontLeft or backRight.** Next, **we showed them how they would write code for both the teleOP period and the Autonomous period by importing a certain package into the code.** Finally, we showed them **how to set powers for their motors.**"

Motivating Hype Nyte

our community

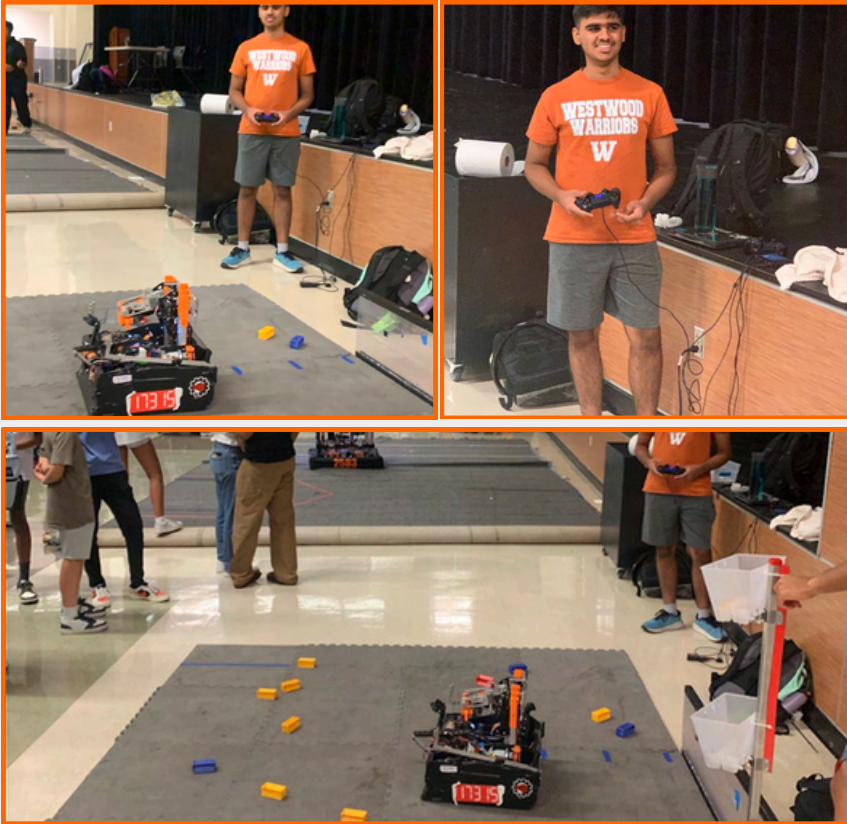


Goals:

To help raise interest in robotics in our High School and also to give a sort of introduction to FTC for those in our school who are hoping to pursue it this season.

WHO:

Freshmen and other potential organization members from our school with an interest in robotics.



TEAM TAKEAWAYS:

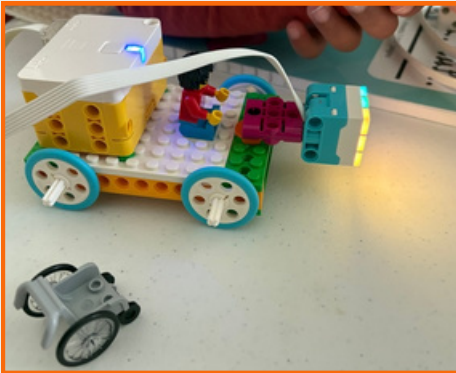
Deekshita - "Going to Hype Nite was such a fun time! We started off with driving our robot around for the freshmen to see how FTC works. One of my favorite parts of this event is that we hand out donuts to all the students who show up and ask questions. Many of the kids were really excited to be there and asked many different questions. One of them **who I remember was named Ishaan. Ishaan was so excited looking at the Tomahawk robot and was wondering how the robot controls worked.** After I explained the basic controls, **he asked to drive the robot. Ishaan told me that he had wanted to join Westwood robotics for a long time** and thanked me for the demo. I really like this experience because **it showed me how something as simple as a robot demo can inspire others to learn more about robotics."**

Ritvij - "Besides showing the freshmen the robot, we also taught those who stuck around about aspects of the robot and design. My job was to teach design. **I started off by showing a video of the Into the Deep season release** and asking the club freshmen on their thoughts on the video, along with any sort of ideas they had to be able to accurately intake and deposit the game elements rapidly. **This was meant to mostly improve their ability to brainstorm ideas to make them be able to plan more efficiently when Decode would roll around.** After the video, **I showed them the different types of parts we use including servos, motors, and other servo attachments such as servo horns and servo gears."**

Motivating

our community

Spike Lego Tutoring



Goals:

To help teach kids basics of robotics and to give a sort of introduction to FLL for those in our school.

WHO:

Elementary students that have an interest in STEM and want learn robotics

TEAM TAKEAWAYS:

Daniel - "We taught Barakh about basic programming and building of the Spike Lego Prime set **every week**. It was the process of building, finding mistakes, innovating and redesigning the build that made this process **meaningful**. Additionally, we taught him basic **programming language using code blocks** such as if conditions and loops. Everytime we visit, Barakh learns something new and has an amazing experience working with us.

Isaac - "Tutoring Barakh reminds me what is really important in FTC: learning about real world STEM skills. I love being a part of his journey as he slowly learns more and more about programming and engineering design. Over the span over 5 weekends, **he has learned fundamental programming concepts** such as conditionals, motor controls, and loops in the SPIKE coding interface. **He has also started learning CAD in onshape**, and designed basic shapes. The shine in his eyes when he understands something new, and his passion to learn more are truly inspiring. "

Sanaika - "Although the journey of teaching Veda has just begun, we have already **gathered countless meaningful and beautiful experiences** that are guiding her toward becoming a true passionate robotics thinker by meeting **weekly**. By working with the **LEGO Spike kit**, she had developed a **natural understanding of the essential components** of a robot and the overall design process. Additionally, **Scratch, a beginner-friendly, block-based programming platform**, has provided her with a strong foundation in **basic coding concepts(loops, variables)** and execution, enabling her to successfully start to **learn on how to program and run a robot.**"

Motivating

our community

Diwali at the Domain



Goals:

To help teach kids basics of robotics, what we do as part of FTC. As well as to have an interactive demo to help them spark their interest, as well as to guide them on how to get involved.

WHO:

The overall Community in the North Austin, Domain area.



TEAM TAKEAWAYS:

Jusuf - "Seeing the children's faces light up and their curiosity spark as they got to play with the ankle biter robot was truly inspiring! **I primarily helped with our robot demonstration, where we talked to interested kids, parents, and even robotics alumni about what we do**, what they can do to be more involved in robotics."

Minessh - "Going to the Diwali event was a great experience. One fond memory I have was meeting Mr. Vidhesh. **Mr. Vidhesh was one of the many people to come to our pit**. However, he listened to our story and our goals. He explained how he was proud of us and wanted too help out. **We told him about our major outreach initiatives in Dubai and the Philippines**. He also told other about our initiatives at the event leading more people to come to our stalls. **This taught me that many in the community are interested in STEM and robotics as a whole and we have to find them and tell them about ourselves to make an impact.**"

Prahastha - "The Diwali event was the best community outreach we have done so far. Talking with people, demonstrating to kids, and explaining to them the world of FIRST was truly fascinating. I mainly contacted the organization, drafted the plan and layout for the event, and helped coordinate the details. **It was really exciting to set up our stall at such a large event, where we got to introduce FIRST to many kids and parents**. We were thrilled to discover that most of the kids were already part of the FIRST community through FIRST Lego Robotics."

Motivating

our community

Anderson Mill Carnival



Goals:

To help teach kids basics of robotics, what we do as part of FTC. As well as to have an interactive demo to help them spark their interest, as well as to guide them on how to get involved.

WHO:

The overall Community in the Anderson Mill area.



TEAM TAKEAWAYS:

Daniel - "It was really fun explaining robot to kids and see their face spark with interest. I mainly helped set up the field. There was a lot of adults who were interested in robotics and we introduced it to them and we had hands on activity by **letting kids drive our robot**. It was a great experience to come to this carnival and introduce something unique and exciting for our community."

Arjun - "Going to the Anderson Mill PTA Carnival was truly a precious moment as I saw the **children thrilled to drive the bot**. I mainly helped set up the drive bot, along with setting up the booth, and occasionally presented our old bot to the parents. The **festival overall was very inviting** and the **parents and children seemed to be interested in pursuing robotics** as they **asked many questions** about how we manage to build big robots and how we successfully compete. The kids and parents were particularly **intrigued and surprised by our bot from Into the Deep**, as they were thrilled to see the full extent of the bot and how we were able to build and code a bot of that scale in a short amount of time. "

Motivating FLL 66670 Mentoring

our community



CHALLENGE

Goals:

Mentor a local FLL team who reached out asking for assistance with the creation of their bot, helping give them advice on their innovation project and presentation as well as any sorts of general advice to make the team succeed.

WHO:

FLL team 66670 Spicewood Superstars, from Spicewood Elementary School



TEAM TAKEAWAYS:

Ritvij - "Mentoring Spicewood Superstars weekly was such a great experience! One of my favorite parts about mentoring them was how much it **taught me about teaching others**. Many times they would get angry when things didn't work or **didn't have the motivation to continue to work**. However, by mentoring throughout the season, I learned not only how to lead but how to teach. As an **FLL Alumni**, it was also fun to go back and help other compete in the game that I used to compete in. At the start, I found it difficult. Although this was my second year mentoring for FLL, **it still felt hard getting used to considering many of the robot strategies had changed from when I competed**. However, I still enjoyed learning about the new mechanisms and strategies other teams used and helping them do their best."

Motivating

our community

Project Philippines FTC



Goals:

Continuing on our goals to spread FIRST, the Our Lady of Joy Learning Center asked us how they could get involved in robotics and asked us to help them through the logistics of doing FTC.

WHO:

Team PHILBOT26 31786 from OLJLC in Cebu

TEAM TAKEAWAYS:

Deekshita - "When we decided to create the first FTC team in the Philippines, we heavily underestimated the number of challenges we would have, whether it be time zones or typhoons. However, we're incredibly grateful to have collaborated with FRC team Tsinelas who offered assistance on shipping parts in the Philippines, FTC team Clueless who helped mentoring, and the numerous sponsors who chipped in to sent them to compete in Vietnam. I hope these kids will gain many memories from these experiences and explore STEM further in the future."

Prahastha - "Starting the first robotics team in the Philippines was much harder than we expected, but we've kept pushing forward. We're incredibly grateful for the support and collaboration from Team Clueless and the first FRC team, Tsinelas. Along the way, **we've faced challenges like part delays, high costs, and even typhoons**—but we're still standing and moving ahead."

Ritvij - "Creating the first FTC team in the Philippines was a major challenge that required extensive collaboration. **Team 11212 Clueless, a two-time Inspire Award winner at Worlds and founders of the Global Mentorship and Ignite Robotics programs**, helped significantly by supporting paperwork and mentoring. We also received key mentorship from the **University of CITU**, which **provided meeting space**, and from Team 9715 Tsinelas, the only FRC team in the Philippines, who offered valuable in-person support to Philbots."

Motivating Project Dune

our community



Goals:

Assist in creating a new league in the UAE Area to give more students the chance to compete in the FIRST Tech Challenge. Besides just funding the creation of the league, assist in mentoring the newly created teams and helping the overall region run smoothly.

WHO:

FIRST in UAE with UWR Robotics and 23400 UForce



TEAM TAKEAWAYS:

Ritvij - "Project Dune was one of my favorite projects this entire year. We helped advocate for funding and used here in the US and donated them to the project. Although overall the project was projected to cost almost **30,000** dollars, due to having multiple partners such as **UForce and Unique World Robotics** who helped raise a majority of the funds in the UAE. However, we were able to still able to contribute to the project by offering mentorship to some of the teams who were created there and by advocating for funds. One issue that we found quickly was the **difficulty in fundraising from simply events alone**, we thus decided to also fundraise from this initiative from our **Community Events**. "

Arjun - "Participating in Project Dune was such a great opportunity. One of the challenges we faced similar to the Philippines was **getting permissions for our parts to pass customs**. Since we had a contact in 23400 UForce on the other side in the UAE. Thus although one of our **packages was kept back**, UForce was eventually able to recover them and **ensure they could be distributed out**."

Jusuf - "One of my favorite memories in Project Dune was helping raise funds for the project. Since it was **difficult for UForce to raise money alone**, they collaborated with us and Pro2type to be able to be able to earn enough. Also, an issue we realized was that by starting the program, many **teams would be left without proper guidance** leading them to **eventually lose interest** which we had seen before in our previous years of FTC. To counter this, **we offered help wherever we could** and whenever they needed it along with Unique World Robotics and UForce.

Motivating

our community

FLL Mentoring 73061 Spartabots



Goals:

Help out a nearby FLL team execute their missions weekly. Also to help guide them through mock judging for awards preparation as well as to help give ideas on how to execute their innovation project.

WHO:

FLL team 76061 Spartabots

TEAM TAKEAWAYS:

Jusuf - "Working with the team and seeing all of the young minds create was truly inspiring. We provided numerous **helpful comments on their presentation, portfolio, robot design, and code**. I mainly helped with their presentation and portfolio, suggesting they read the judge feedback from their **mock presentation**, use fewer words and more bullet points in their portfolios, and speak more during their presentation."

Karthik - "Tutoring the FLL team provided my with a valuable experience in mentoring and sharing my knowledge. We helped the students **identify effective strategies** and the best order for completing the missions ensuring success. By supporting them in coding by explaining and implementing effeicient and neat stragies; and also suggesting hardware changes so it doesnt make the code complicated. We strenghted our communication and problem-solving skills as a team. This oppurtunity not only allowed us to guide younger students but also taught us the importance of collaboration and adabtality in FIRST outreach."

Daniel - "It was a remarkable experience working with a team! We gave lots of useful feedback to the team's presentation and portfolio while also helping them understand problems in their code. One lesson that I taught to the team was that everyone should understand **how the basic setup works**, that way it isn't just one person working on the code, and one person working on the presentation. **Within an hour, I could see the team's precision increase and managed to solve two more missions.**"

Motivating Minibots

our community



Goals:

Guide elementary students through fun engaging STEM activities and exercise thinking skills to help them slowly develop an interest in Robotics and potentially give them the apssion to join the club eventually.

WHO:

Elementary students in the Westwood area



TEAM TAKEAWAYS:

Daniel - "Being a counselor at minibots was a an awesome experience. Minibots is an outreach event hosted by us where we lead elementary students to work on hands-on STEM activities similar to robocamp to further boost interest in the idea of robotics in our local community. The kids had lots of fun building Reptile robots, a kit, and had a great learning experience."

Sanaika- "It was a wonderful experience to mentor younger kids and watch them expand thier knowledge on something we all loved in that room. We had built a strong drivetrain which was able to place as the winner in more than 2 rounds. Watching the kids learn and brighten up from their achievements motivated me and them overall. At the end of the day, we had a productive day in which we had successfully taught younger kids to achieve their goals through robotics."

Jusuf- "It was a wonderful experience to mentor younger kids and watch them expand thier knowledge on something we all loved in that room. We had built a strong drivetrain which was able to place as the winner in more than 2 rounds. Watching the kids learn and brighten up from their achievements motivated me and them overall. At the end of the day, we had a productive day in which we had successfully taught younger kids to achieve their goals through robotics."

Motivating Mechanical Maestros 28784N Mentoring

our community

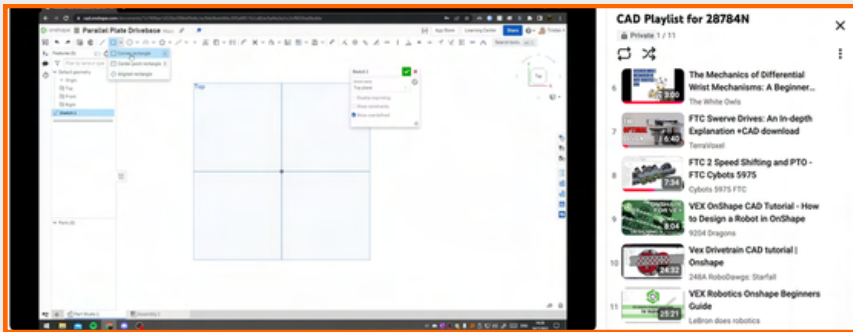


Goals:

To help a VEX team weekly in our area learn how to cad and overall learn how to better design their robot mechanically to work more consistently by using design softwares such as Onshape.

WHO:

VEX Team 28784N
Mechanical Maestros



TEAM TAKEAWAYS:

Ritvij - "Mentoring Mechanical Maestros was such a great experience. The team was based from Canyon Vista Middle School and had experience previously from FLL. As a roboticist who had never officially done VEX before, It was hard to be able to adjust at the start with the robotics that VEX had. One thing that I noticed off the bat was that VEX had zero custom parts allowed which I felt limited the creativity of possible solutions. However, I still tried to **teach what I knew about CAD** and helped **fix any issues they had** when they started using Onshape for the first time."

Shrikar - "Mentoring the Mechanical Maestros was such a great opportunity. As someone who does VEX Robotics from beforehand, it was a great opportunity for me to spread what I had already done well. One issue that I remember they were dealing with was fully designing their robot out in a software before building it. The benefits of this is that you make **many less hardware mistakes and can account for the majority of your hardware bugs by having a 3D Model to follow**. We also taught them how to efficiently build a drivetrain and an intake for this years game. **We mainly focused on how to build a bot that could be competitive and easy to build**. We also focused on build quality and how to build a bot that would survive a vex v5 competition."

Motivating

our community

CVMS STEM Night



Goals:

To go to a middle school to spread our name along with telling students there about our schools robotics program to gain interest for potential future members of our team and our club.

WHO:

CVMS Robotics Club and School

TEAM TAKEAWAYS:

Aron - "Attending and presenting at CVMS STEM night was a great experience. One of my favorite parts was **showing the kids our Into the Deep robot from the season prior** and explaining how we were able to successfully iterate throughout the year. The kids were very interested and **asked many questions about the robot** which I was glad to see. I hope that in the future, our efforts will lead more students from these feeder robotics program **to have interest** in joining our team and program leading to more **sustainability of talent and people to ensure the team survives.** "

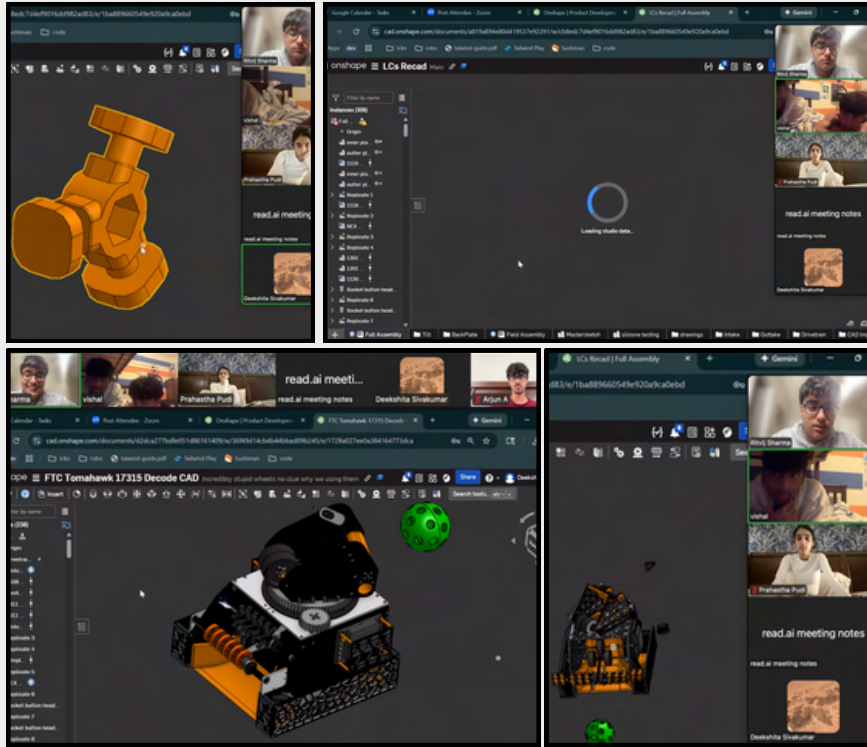
Isaac - "It was definitively a learning experience for me as well as them. **Organizing an event primarily by ourselves was a challenge**, but I think this experience is really helpful for us if we organize another event in the future. The kids themselves were **interested to learn about what we do and how our robot worked, and it was cool to see their original ideas.** As a former student of this middle school, I really enjoyed giving back to the community. I hope this event taught them a lot, and that they have a greater chance to **pursue robotics at my high school** and maybe give back like I did. "

Jusuf - "Presenting at CVMS STEM night was an amazing experience. I mainly showed and explained **our robot from last season**, walking the students through how our **intake and outtake mechanisms** work. Seeing all the kids that gathered to listen to us was truly inspiring. Many of them **asked questions about our robot and FTC** as well. It was clear they were genuinely interested, and I hope that interest will carry on to when they enter high school and join robotics, just as we did."

Motivating

our community

24167 Orion Mentoring



Goals:

To help an FTC team learn how our own robot worked and how we were able to succeed at creating a robot that was functional and worked consistently. Also, to give advice to them about how to make a consistently working robot that would score high.

WHO:

24167 Orion

TEAM TAKEAWAYS:

Ritvij - "Mentoring Orion was a great experience for us. It showed me that engineering support isn't just about CAD or mechanisms but it's about communication, adaptability, and patience. Since we majorly mentored them solely over zoom, **it was harder for us to help them with their own robots concerns.** However, mentoring them taught me about how to stay patient and **adjust how we would share resources and teach.**

Prahastha - "Working with Orion was a great opportunity. **One issue that we found they were facing was limited funding reinforced the importance of designing with accessibility in mind.** We told them it was important to Explore **alternatives like surgical or latex tubing** and emphasized durable 3D printing practices highlighted how thoughtful material choices can maintain performance while staying within budget."

Arjun - "The experience underscored how **collaborative problem-solving strengthens outcomes on both sides of mentorship.** By offering CAD guidance, design feedback, and build strategies while also learning from Orion's challenges, we **expanded our reach as mentors and contributed to a more supportive and connected engineering community.**"

Motivating

our community

17113 Hunga Munga Mentoring



Goals:

Helping a sister FTC team with less experience learn how to better present the information for their portfolio as well as general advice for portfolio. Also, to give feedback for their presentation and help their presentation be the best it can be.

WHO:

17113 Hunga Munga

TEAM TAKEAWAYS:

Ritvij - "Hunga Munga was our Westwood Robotics sister team. Since this was the majority of their first year, we helped them with many concepts. One of these being the portfolio itself. Some parts that we helped them with was **blocking and structuring text so it was readable yet informative**. The way we did this was suggesting how to **package their text better on the portfolio to be more aesthetically pleasing**. We also helped them identify which award they wanted to win and **help them plan on how to hit the criteria well for each of the requirements for that award**. For Hunga Munga, they wanted to go for the Think Award, so we suggested them to include calculations and design process material to help better present their design journey."

Deekshita - "We also helped Hunga Munga whenever they had issues with their robot. Hunga Munga started out with a **starter bot and slowly iterated into a custom robot**. One of the issues they faced was with their **rubber band intake which could break frequently**. We suggested to them to play more cautiously in their games if they chose to keep that type of intake. **We also taught them the importance of ranking points from a strategic perspective explaining how this year winning wasn't everything** and that getting these ranking points every game would choose who would be ranked high vs. low which could be the difference between advancing and not advancing."

Motivating

our community

18886 Atlatl Mentoring

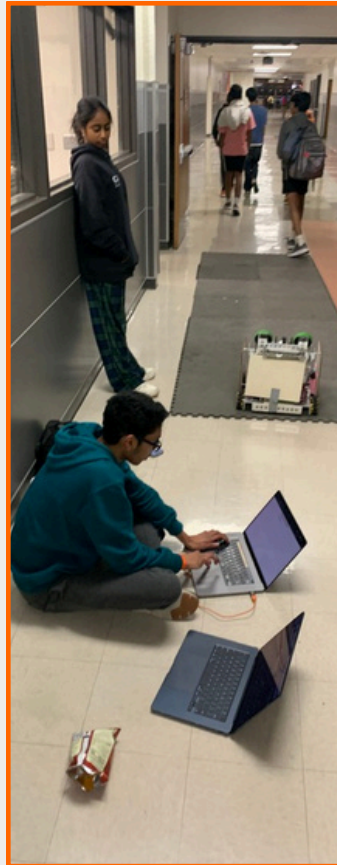


Goals:

To help an FTC team learn how our own robot worked and how we were able to succeed at creating a robot that was functional and worked consistently. Also, to assist their presentation and portfolio preparation.

WHO:

18886 Atlatl



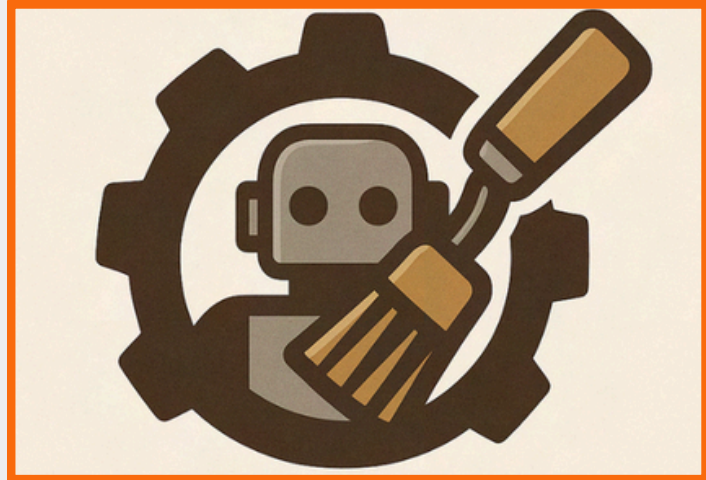
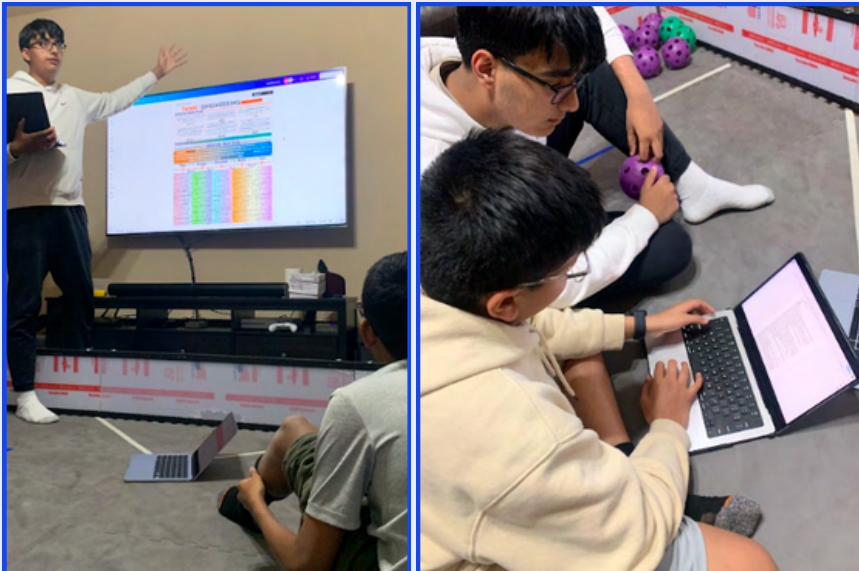
TEAM TAKEAWAYS:

Ritvij - "We mentored 18886 Atlatl throughout the season. Atlatl is another one of our sister teams from Westwood. **Atlatl was also a complete rookie team with zero FIRST experience beforehand.** One of the issues that they faced was making a consistent robot, an issue they faced throughout the year. One way we helped alleviate this was **encouraging them to follow a blueprint that has been shown to work** before switching to their own creative idea. **Because of this, Atlatl decided to make a robot similar to the ones in VEX** that had been proven to work well before to ensure that their robot would be consistent and work decently well. We also gave Atlatl help with their portfolio offering them feedback on the structure of their portfolio as well as their presentation to ensure they could clearly hint to the judges which award they were going for."

Aditya - "Another way that we helped Atlatl throughout the year was with their software. One of the main issues that Atlatl faced from the start of the year was the lack of software experienced people on their team. Usually **most rookie teams had members who had some coding background. However, their team didn't have any which made it a challenge for them to write code.** One of the issues they were facing was learning how to tune their roadrunner for their version 2 robot. We thus decided to help them, **leading with demo showing them how to tune**, before stepping back and **watching them tune themselves giving advice where we could to ensure they would retain this skill.**"

Motivating 30358 Mentoring

our community



Goals:

Help give advice to a rookie team about how to better build their robot to score more and be consistent. Also helping a rookie team develop their portfolio and presentation and teaching them how to aim for different types of awards.

WHO:

30358 Gearchaeologists



TEAM TAKEAWAYS:

Ritvij - "Mentoring Gearchaeologists was a great experience. One of my favorite experiences over the course of the season was watching them grow and improve over the course of the season. When I started to mentor them, they had many **issues with their robot such as consistency with their hardware, a lack of teamwork, and no clear direction or plan for the season besides simply continuing with their GoBilda starter bot.** However, over the course of the season, they slowly improved getting better at working together and learning how to fix their hardware issues and eventually having a plan. Another part of FTC that I helped them with was their portfolio, something that they were struggling with at the start since they didn't understand how the criteria for them worked. I then gave a presentation to them about **how the portfolio works and how they should approach awards suggesting they attempt to work towards a technical award after they told me they had documented their season journey really well in a sort of engineering notebook.** I also continued to help them by constantly giving them feedback on their portfolio regularly helping them know what they were doing right and what they needed to improve upon."

Motivating Orbit Mentoring

our community



Goals:

Ensure the continuity of Project Dune by mentoring some of the teams that we helped finance to create to ensure the team will survive allowing the UAE region to survive for years to come.



WHO:

28831 Orbit as part of Project Dune

TEAM TAKEAWAYS:

Ritvij - "Orbit was one of the teams where the majority of our parts had gone to. Since we wanted to ensure they would be successful for years to come and not lose interest, we began to mentor them throughout the season. We would give them advice on how to use **3D designing softwares and 3D printed parts** in combination with Gobilda and other vendors standardized parts to make an efficient robot. One of the main issues they faced early in the season was a stable and consistent robot. **We suggested that there were many small aspects they could fix by just paying attention to detail which would overall make them better.** Some of these small details were the **friction on bearings for the shooter or the usage of thread lockers such as locktite.** Overall, it was difficult to mentor throughout the year due to the distance. The majority of our **mentorship would occur over text making it difficult to fully diagnose issues** when they had them. But, we learned a **great amount about how to mentor even when you aren't physically somewhere.**"

Motivating AMD Mentoring

our community

Goals:

Ensure the continuity of Project Dune by mentoring some of the teams that we helped finance to create to ensure the team will survive allowing the UAE region to survive for years to come.

WHO:

28588 AMD as part of Project Dune



TEAM TAKEAWAYS:

Ritvij - "Along with Orbit, we also mentored team 28588 AMD throughout the year too. AMD was a relatively new team and was made in the offseason. **They still however needed help learning how to design for a shooter game this year.** After we sent them parts, we **attempted to explain to them how to create flywheel geometry and suggested to them to create a starter bot** for the first half of the season. Our reasoning for this was that by having a starter bot, **they could have a reliable base which would have a stable floor in terms of their scoring and their ranking points while allowing their software team to start programming the robot while the hardware team could learn how to create a more advanced robot.** Although they initially opposed the idea, additional feedback from both UForce and United World Robotics made them realize this was the best strategy. **AMD also used many 3D printed parts on their robot to ensure structural stability which was important especially since they had limited access to parts due to Gobilda's lengthy shipping times.**"